

Origin of Mass and the Higgs boson

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Standard Model

▷ Gauge theory of interactions between fermions mediated by vector bosons

- Electroweak interaction: $SU(2) \times U(1)$
- Quantum chromodynamics (strong) interaction: $SU(3)$
- No gravitation

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i\bar{\psi} \not{D} \psi + h.c.$$

▷ Electroweak Lagrangian

$$\mathcal{L}_{\text{symm}} = -\frac{1}{4} \sum_{A=1}^3 F_{\mu\nu}^A F^{A\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} + \bar{\psi}_L i\gamma^\mu D_\mu \psi_L + \bar{\psi}_R i\gamma^\mu D_\mu \psi_R .$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$$

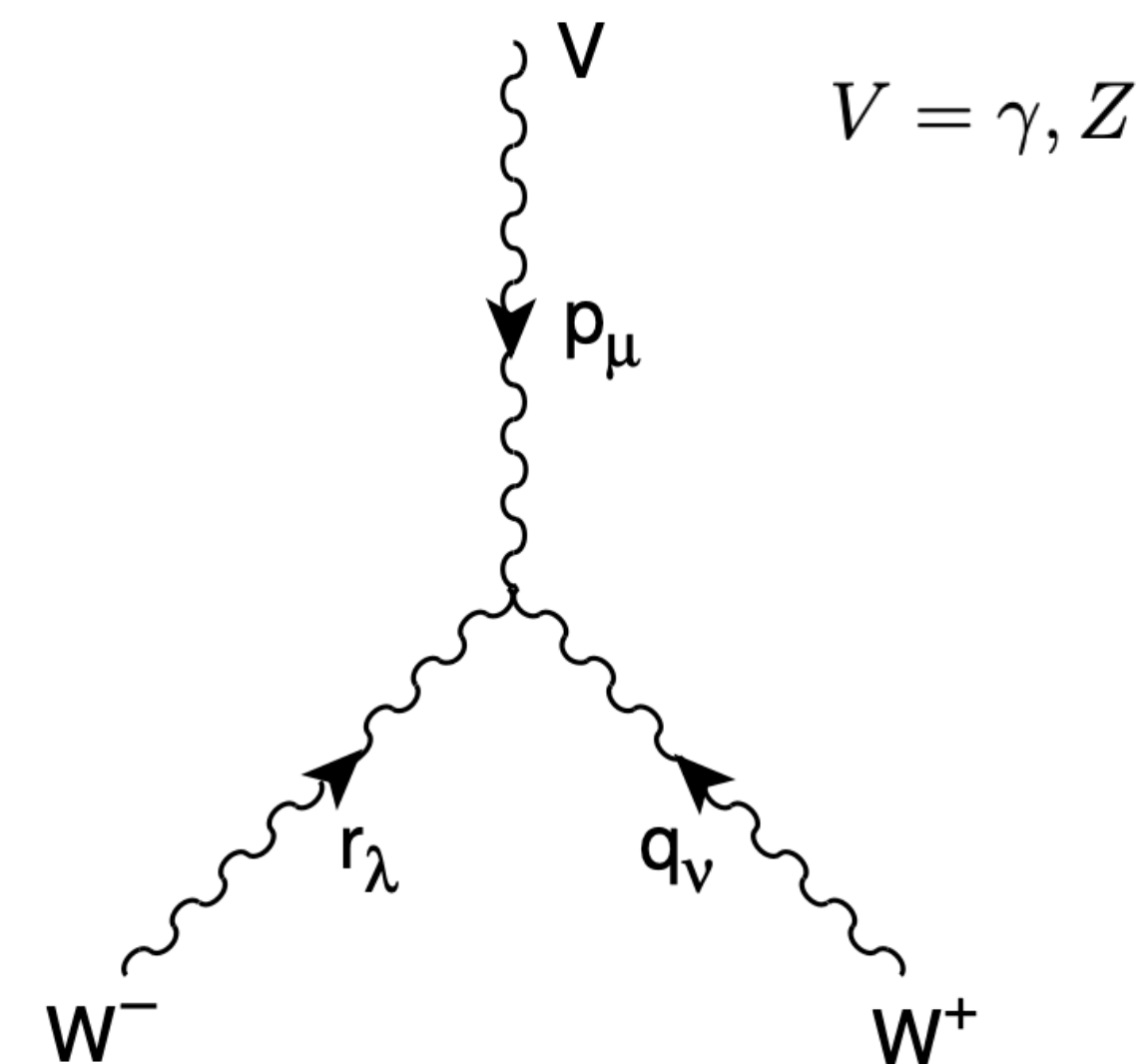
$$F_{\mu\nu}^A = \partial_\mu W_\nu^A - \partial_\nu W_\mu^A - g\epsilon_{ABC} W_\mu^B W_\nu^C$$

$$\psi_{L,R} = [(1 \mp \gamma_5)/2]\psi, \quad \bar{\psi}_{L,R} = \bar{\psi}[(1 \pm \gamma_5)/2]$$

$$D_\mu \psi_{L,R} = \left[\partial_\mu + ig \sum_{A=1}^3 t_{L,R}^A W_\mu^A + ig' \frac{1}{2} Y_{L,R} B_\mu \right] \psi_{L,R}$$

$$A_\mu = \cos \theta_W B_\mu + \sin \theta_W W_\mu^3 ,$$

$$Z_\mu = -\sin \theta_W B_\mu + \cos \theta_W W_\mu^3 .$$



$$g_{W^-W^+\gamma} = g \sin \theta_W = e \quad \text{and} \quad g_{W^-W^+Z} = g \cos \theta_W$$

No mass for fermions and bosons!

▷ All fermions and vector boson mediators are massless

▷ Mass-term for fermions from Dirac equation:

$$-m\bar{\psi}\psi = -m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R)$$

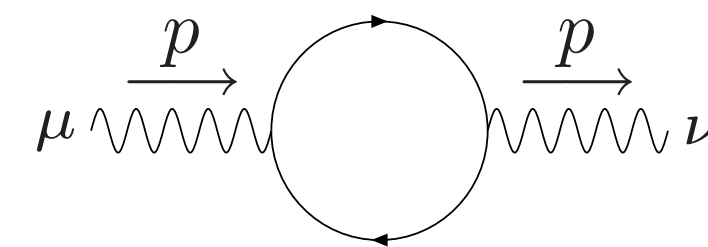
▷ Such a term mixes left-handed and right-handed spinors and is inconsistent with the chiral SU(2) symmetry

Observed particles are massive!

- ▷ All charged leptons are massive
- ▷ Evidence also for quarks to be massive
- ▷ Discovery of W and Z in 1983 proved them to be pretty heavy
- ▷ Need a mechanism to **dynamically generate mass for all particles and conserve the gauge invariance**
- ▷ *Higgs mechanism and goldstone bosons address this problem*
- ▷ Experimental evidence needed to validate theory
 - Predict Higgs production processes and decay rates
- ▷ Direct search
 - produce Higgs and look for decay products: invariant mass and characteristic kinematics
- ▷ Indirect search
 - Calculate contribution of Higgs in corrections to well known SM processes
 - Combined analysis of all precisely measured Z-pole observables and check validity with different Higgs mass assumptions

Generating Mass for W and Z

- ▷ Propagator for massless vector bosons at tree level

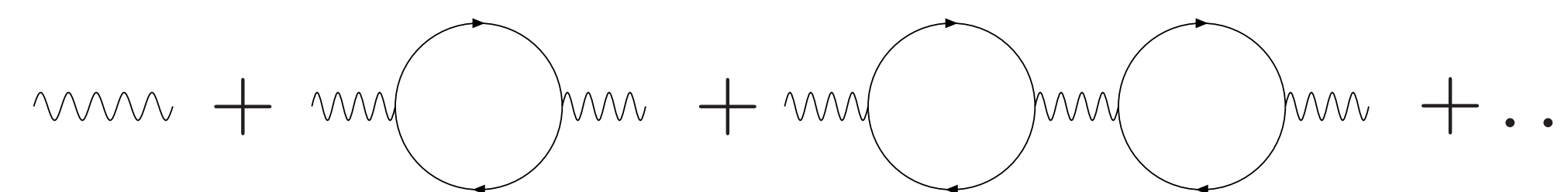


$$i \Pi_{\mu\nu}(p) \equiv i(p_\mu p_\nu - p^2 g_{\mu\nu}) \Pi(p^2)$$

Gauge invariance

$$p^\mu \Pi_{\mu\nu}(p) = p^\nu \Pi_{\mu\nu}(p) = 0$$

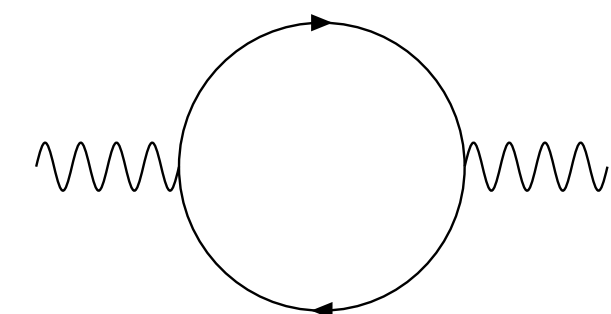
- ▷ Renormalized propagator at higher order



$$= \frac{-i(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2})}{p^2[1 + \Pi(p^2)]}$$

- ▷ For a **massless scalar particle** in the loop

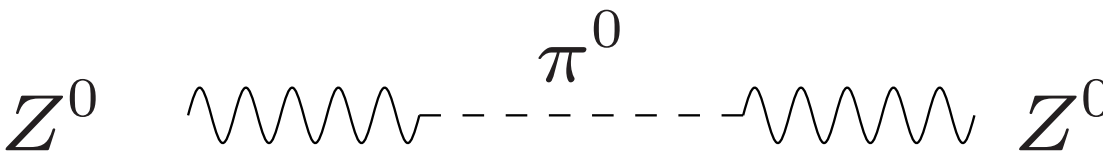
- ▷ For $p=0$ we have our vector boson with $m = gv$!



$$\Pi(p^2) \underset{p^2 \rightarrow 0}{\simeq} \frac{-g^2 v^2}{p^2} \longrightarrow p^2[1 + \Pi(p^2)] = p^2 - g^2 v^2$$

- ▷ Do we have already any such scalar?

Generating W and Z Mass with π^0

- ▷ W and Z couple to charge and neutral currents $\mathcal{L}_{\text{int}} = g_Z j_\mu^Z Z^\mu + g_W (j_\mu^W W^{+\mu} + \text{h.c.})$
- ▷ Charged and neutral pions have very small mass (compared to other hadrons) and are massless in the limit of massless quarks!
- ▷ Pions in the loop could provide the needed mass term  $\Pi(p^2) = -g_Z^2 f_\pi^2 / p^2$
- ▷ Prediction for W and Z mass $m_Z = g_Z f_\pi \quad m_W = g_W f_\pi$
- ▷ Pion decay constant measured from decay rate to be $f_\pi = 93 \text{ MeV}$
- ▷ W and Z mass related in SM through $\frac{m_W}{m_Z} = \frac{g_W}{g_Z} \equiv \cos \theta_W \simeq 0.88$

 - Good agreement with measurement
- We know $g_Z \simeq 0.37$ which corresponds to $m_Z = 35 \text{ MeV}$

 - Off by 3 orders of magnitude!
- Need a new scalar for dynamic mass generation

Higgs field

- ▷ Introduction of a new SU(2) doublet of complex Higgs field

Higgs field $\phi \equiv \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix}$

Higgs potential $V(\phi^\dagger \phi) = -\frac{1}{2}\mu^2 \phi^\dagger \phi + \frac{1}{4}\lambda(\phi^\dagger \phi)^2$

- ▷ Spontaneous breaking of the gauge symmetry to provide mass-like term to vector bosons
- ▷ Coupling of Higgs field to fermions provides mass-like terms and it is often said that “it dynamically generates the fermion mass”

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \phi)^\dagger (D^\mu \phi) - V(\phi^\dagger \phi) - \bar{\psi}_L \Gamma \psi_R \phi - \bar{\psi}_R \Gamma^\dagger \psi_L \phi^\dagger$$

Higgs coupling to fermions ~ mass term for fermions

Spontaneous Symmetry Breaking

▷ New complex SU(2) doublet with four degrees of freedom (massless scalar particles)

▷ Self-interaction potential

$$V(\phi^\dagger \phi) = -\frac{1}{2}\mu^2 \phi^\dagger \phi + \frac{1}{4}\lambda(\phi^\dagger \phi)^2$$

▷ Non-zero vacuum expectation value (VEV) at minimum generates one massive scalar
– remaining 3 scalar absorbed by W, Z, and photon

$$\Phi^\dagger \Phi = \frac{v^2}{2}$$

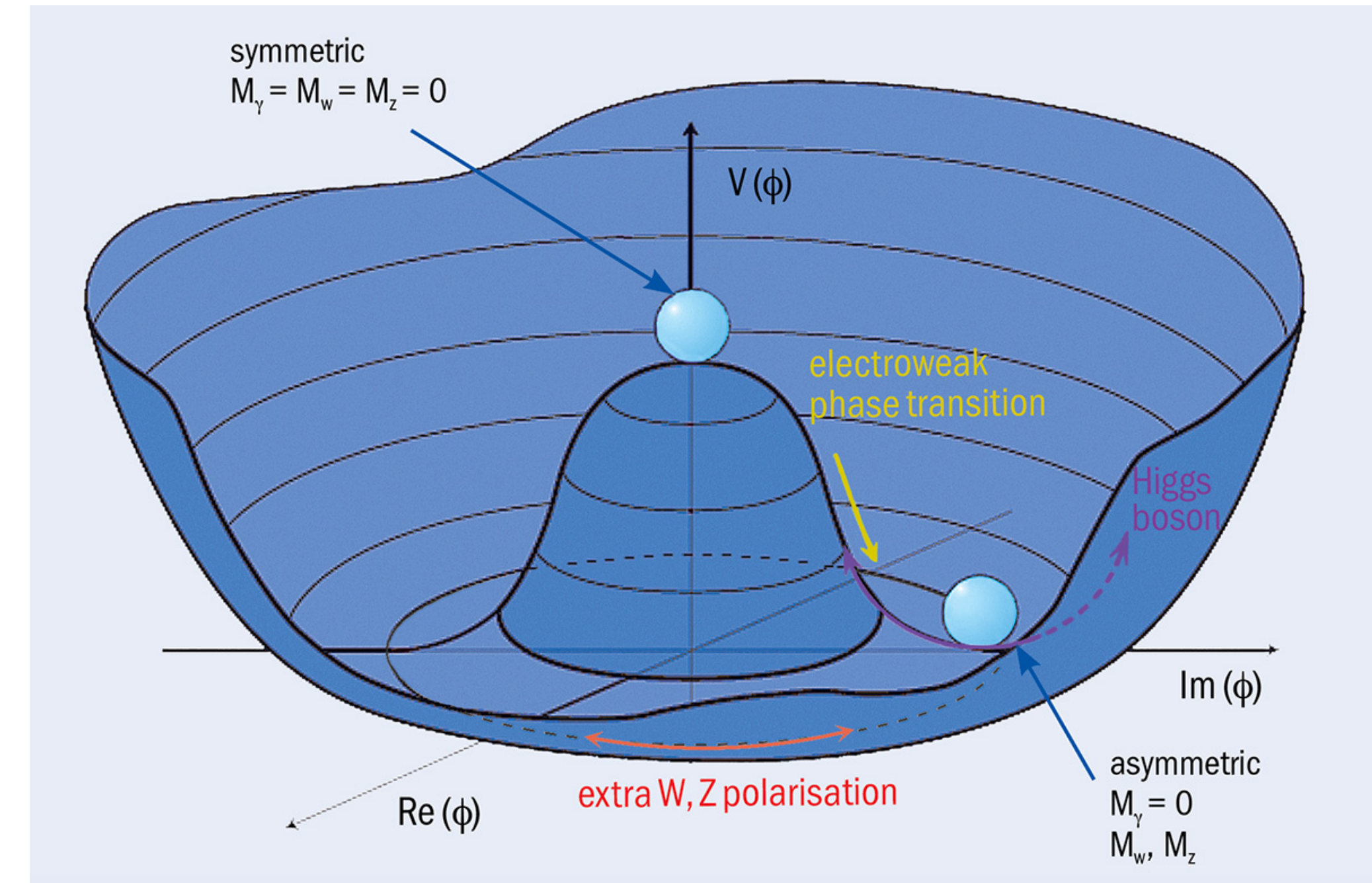
▷ The coupling of this scalar must generate correct mass

▷ Therefore the VEV must be $v = 246 \text{ GeV}$ $m_Z = g_Z v \simeq 91 \text{ GeV}$

▷ New massive scalar particle commonly known as Higgs Boson

– Interactions and decay rates known precisely as a function of its mass

– But its **mass not fixed and must be measured**



$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

$$V(h) = \text{const} + \lambda v^2 h^2 + \lambda v h^3 + \frac{\lambda}{4} h^4$$

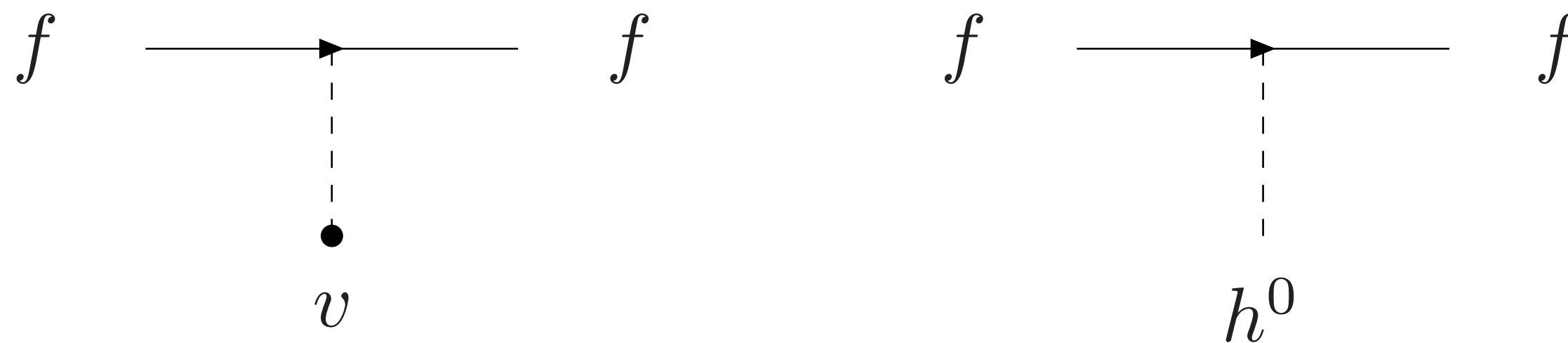
$$m_H^2 = 2\lambda v^2$$

Fermion Mass thanks to Higgs

- ▷ In addition to Z and W, also fermions acquire mass in SM thanks to Yukawa coupling to Higgs boson

$$\mathcal{L}_{\text{Yukawa}} = -h_u(\bar{u}_R u_L \Phi^0 - \bar{u}_R d_L \Phi^+) - h_d(\bar{d}_R d_L \Phi^{0*} + \bar{d}_R u_L \Phi^-) + \text{h.c.}$$

$$-m\bar{\psi}\psi = -m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R)$$



$$g_{hf\bar{f}} = m_f/v$$

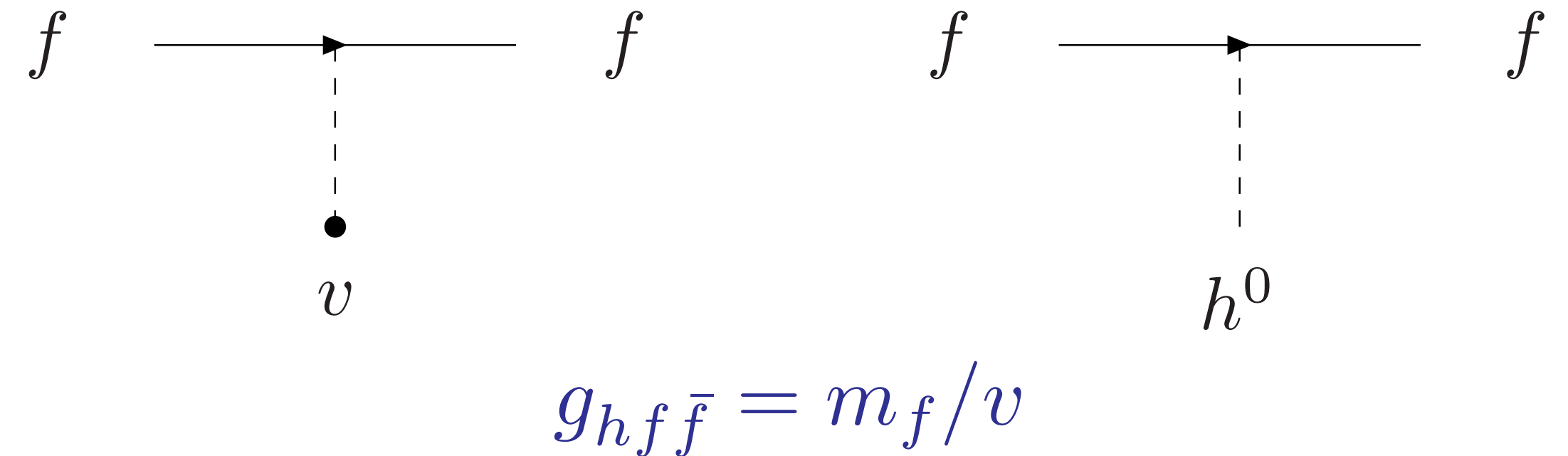
fermions	SU(2)	U(1) _Y
$(\nu, e^-)_L$	2	-1
e_R^-	1	-2
$(u, d)_L$	2	1/3
u_R	1	4/3
d_R	1	-2/3

- ▷ Coupling proportional to fermion mass
 - Will determine Higgs decay branching fractions

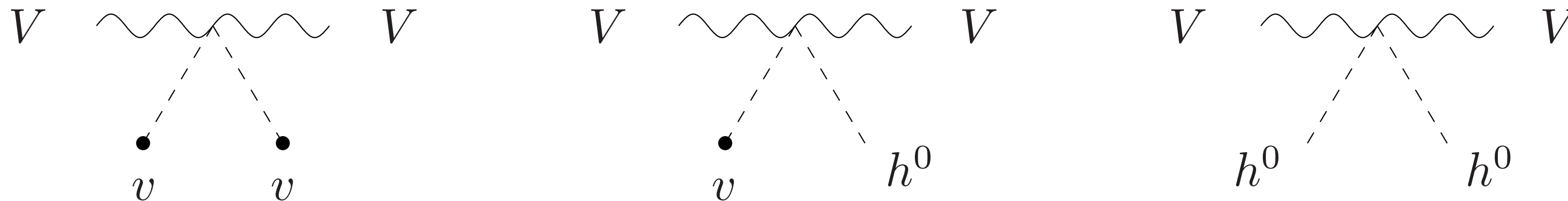
Higgs Coupling

▷ Higgs coupling enhanced for heavier particles

▷ Vector bosons always preferred to fermions
– But must be kinematically allowed



▷ Fixing mass of Higgs fixes decay rates for all final states



$$g_{hVV} = 2m_V^2/v$$

$$g_{hhVV} = 2m_V^2/v^2$$

The Standard Model Lagrangian

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ & + i \bar{\psi} \not{D} \psi + \text{h.c.} \\ & + \bar{\psi}_i Y_{ij} \psi_j \phi + \text{h.c.} \\ & + |D_\mu \phi|^2 - V(\phi)\end{aligned}$$

The complete Standard Model Lagrangian

QCD

Electroweak + Higgs

Fermion interaction
with weak and Higgs

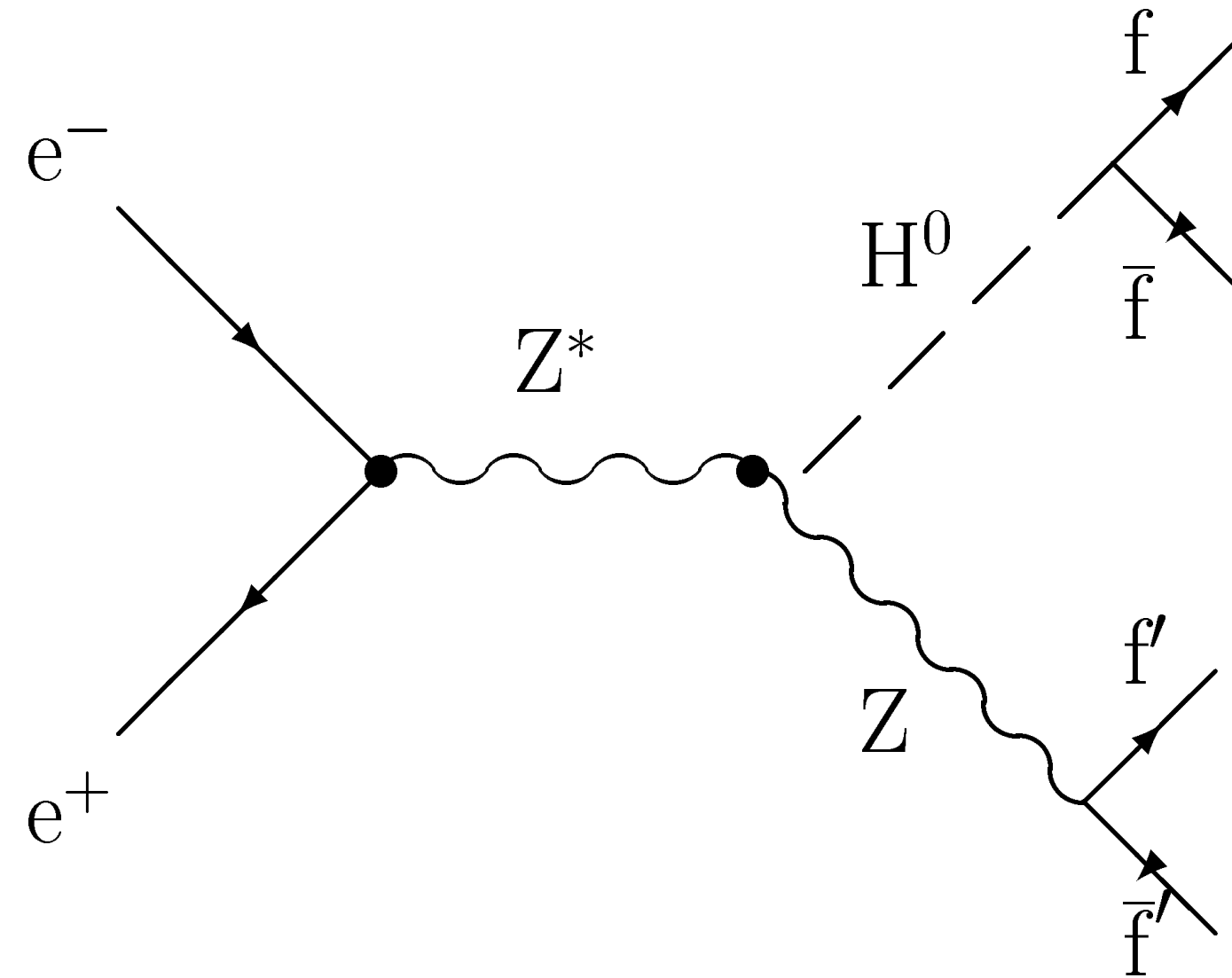
Interaction of matter particles
with Higgs ghosts, virtual artifacts
from the Higgs field

Faddeev-Popov ghosts

$$\begin{aligned}
 & \frac{1}{2} \partial_\nu g_\mu^a \partial_\nu g_\mu^a - g_s f^{abc} \partial_\mu g_\nu^a g_\mu^b g_\nu^c - \frac{1}{4} g_s^2 f^{abc} f^{ade} g_\mu^b g_\nu^c g_\mu^d g_\nu^e + \\
 & \frac{1}{2} i g_s^2 (\bar{q}_i^\sigma \gamma^\mu q_j^\sigma) g_\mu^a + \bar{G}^a \partial^2 G^a + g_s f^{abc} \partial_\mu \bar{G}^a G^b g_\mu^c - \partial_\nu W_\mu^+ \partial_\nu W_\mu^- - \\
 & M^2 W_\mu^+ W_\mu^- - \frac{1}{2} \partial_\nu Z_\mu^0 \partial_\nu Z_\mu^0 - \frac{1}{2 c_w^2} M^2 Z_\mu^0 Z_\mu^0 - \frac{1}{2} \partial_\mu A_\nu \partial_\mu A_\nu - \frac{1}{2} \partial_\mu H \partial_\mu H - \\
 & \frac{1}{2} m_h^2 H^2 - \partial_\mu \phi^+ \partial_\mu \phi^- - M^2 \phi^+ \phi^- - \frac{1}{2} \partial_\mu \phi^0 \partial_\mu \phi^0 - \frac{1}{2 c_w^2} M \phi^0 \phi^0 - \beta_h \left[\frac{2 M^2}{g^2} + \right. \\
 & \left. \frac{2 M}{g} H + \frac{1}{2} (H^2 + \phi^0 \phi^0 + 2 \phi^+ \phi^-) \right] + \frac{2 M^4}{g^2} \alpha_h - i g c_w [\partial_\nu Z_\mu^0 (W_\mu^+ W_\nu^- - \\
 & W_\nu^+ W_\mu^-) - Z_\nu^0 (W_\mu^+ \partial_\nu W_\mu^- - W_\mu^- \partial_\nu W_\mu^+) + Z_\mu^0 (W_\nu^+ \partial_\mu W_\mu^- - \\
 & W_\nu^- \partial_\mu W_\mu^+)] - i g s_w [\partial_\nu A_\mu (W_\mu^+ W_\nu^- - W_\nu^+ W_\mu^-) - A_\nu (W_\mu^+ \partial_\nu W_\mu^- - \\
 & W_\mu^- \partial_\nu W_\mu^+) + A_\mu (W_\nu^+ \partial_\nu W_\mu^- - W_\nu^- \partial_\nu W_\mu^+)] - \frac{1}{2} g^2 W_\mu^+ W_\mu^- W_\nu^+ W_\nu^- + \\
 & \frac{1}{2} g^2 W_\mu^+ W_\nu^- W_\mu^+ W_\nu^- + g^2 c_w^2 (Z_\mu^0 W_\mu^+ Z_\nu^0 W_\nu^- - Z_\mu^0 Z_\mu^0 W_\nu^+ W_\nu^-) + \\
 & g^2 s_w^2 (A_\mu W_\mu^+ A_\nu W_\nu^- - A_\mu A_\mu W_\nu^+ W_\nu^-) + g^2 s_w c_w [A_\mu Z_\nu^0 (W_\mu^+ W_\nu^- - \\
 & W_\nu^+ W_\mu^-) - 2 A_\mu Z_\mu^0 W_\nu^+ W_\nu^-] - g \alpha [H^3 + H \phi^0 \phi^0 + 2 H \phi^+ \phi^-] - \\
 & \frac{1}{8} g^2 \alpha_h [H^4 + (\phi^0)^4 + 4 (\phi^+ \phi^-)^2 + 4 (\phi^0)^2 \phi^+ \phi^- + 4 H^2 \phi^+ \phi^- + 2 (\phi^0)^2 H^2] - \\
 & g M W_\mu^+ W_\mu^- H - \frac{1}{2} g \frac{M}{c_w^2} Z_\mu^0 Z_\mu^0 H - \frac{1}{2} i g [W_\mu^+ (\phi^0 \partial_\mu \phi^- - \phi^- \partial_\mu \phi^0) - \\
 & W_\mu^- (\phi^0 \partial_\mu \phi^+ - \phi^+ \partial_\mu \phi^0)] + \frac{1}{2} g [W_\mu^+ (H \partial_\mu \phi^- - \phi^- \partial_\mu H) - W_\mu^- (H \partial_\mu \phi^+ - \\
 & \phi^+ \partial_\mu H)] + \frac{1}{2} g \frac{1}{c_w} (Z_\mu^0 (H \partial_\mu \phi^0 - \phi^0 \partial_\mu H) - i g \frac{s_w^2}{c_w} M Z_\mu^0 (W_\mu^+ \phi^- - W_\mu^- \phi^+) + \\
 & i g s_w M A_\mu (W_\mu^+ \phi^- - W_\mu^- \phi^+) - i g \frac{1-2c_w^2}{2c_w} Z_\mu^0 (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) + \\
 & i g s_w A_\mu (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) - \frac{1}{4} g^2 W_\mu^+ W_\mu^- [H^2 + (\phi^0)^2 + 2 \phi^+ \phi^-] - \\
 & \frac{1}{4} g^2 \frac{1}{c_w^2} Z_\mu^0 Z_\mu^0 [H^2 + (\phi^0)^2 + 2 (2 s_w^2 - 1)^2 \phi^+ \phi^-] - \frac{1}{2} g^2 \frac{s_w^2}{c_w} Z_\mu^0 \phi^0 (W_\mu^+ \phi^- + \\
 & W_\mu^- \phi^+) - \frac{1}{2} i g^2 \frac{s_w^2}{c_w} Z_\mu^0 H (W_\mu^+ \phi^- - W_\mu^- \phi^+) + \frac{1}{2} g^2 s_w A_\mu \phi^0 (W_\mu^+ \phi^- + \\
 & W_\mu^- \phi^+) + \frac{1}{2} i g^2 s_w A_\mu H (W_\mu^+ \phi^- - W_\mu^- \phi^+) - g^2 \frac{s_w}{c_w} (2 c_w^2 - 1) Z_\mu^0 A_\mu \phi^+ \phi^- - \\
 & g^1 s_w^2 A_\mu A_\mu \phi^+ \phi^- - \bar{e}^\lambda (\gamma \partial + m_e^\lambda) e^\lambda - \bar{\nu}^\lambda \gamma \partial \nu^\lambda - \bar{u}_j^\lambda (\gamma \partial + m_u^\lambda) u_j^\lambda - \\
 & \bar{d}_j^\lambda (\gamma \partial + m_d^\lambda) d_j^\lambda + i g s_w A_\mu [-(\bar{e}^\lambda \gamma^\mu e^\lambda) + \frac{2}{3} (\bar{u}_j^\lambda \gamma^\mu u_j^\lambda) - \frac{1}{3} (\bar{d}_j^\lambda \gamma^\mu d_j^\lambda)] + \\
 & \frac{i g}{4 c_w} Z_\mu^0 [(\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (\bar{e}^\lambda \gamma^\mu (4 s_w^2 - 1 - \gamma^5) e^\lambda) + (\bar{u}_j^\lambda \gamma^\mu (\frac{4}{3} s_w^2 - \\
 & 1 - \gamma^5) u_j^\lambda) + (\bar{d}_j^\lambda \gamma^\mu (1 - \frac{8}{3} s_w^2 - \gamma^5) d_j^\lambda)] + \frac{i g}{2 \sqrt{2}} W_\mu^+ [(\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) e^\lambda) + \\
 & (\bar{u}_j^\lambda \gamma^\mu (1 + \gamma^5) C_{\lambda \kappa} d_j^\kappa)] + \frac{i g}{2 \sqrt{2}} W_\mu^- [(\bar{e}^\lambda \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (\bar{d}_j^\kappa C_{\lambda \kappa}^\dagger \gamma^\mu (1 + \\
 & \gamma^5) u_j^\lambda)] + \frac{i g}{2 \sqrt{2}} \frac{m_e^\lambda}{M} [-\phi^+ (\bar{\nu}^\lambda (1 - \gamma^5) e^\lambda) + \phi^- (\bar{e}^\lambda (1 + \gamma^5) \nu^\lambda)] - \\
 & \frac{g}{2} \frac{m_e^\lambda}{M} [H (\bar{e}^\lambda e^\lambda) + i \phi^0 (\bar{e}^\lambda \gamma^5 e^\lambda)] + \frac{i g}{2 M \sqrt{2}} \phi^+ [-m_d^\kappa (\bar{u}_j^\lambda C_{\lambda \kappa} (1 - \gamma^5) d_j^\kappa) + \\
 & m_u^\lambda (\bar{u}_j^\lambda C_{\lambda \kappa} (1 + \gamma^5) d_j^\kappa) + \frac{i g}{2 M \sqrt{2}} \phi^- [m_d^\lambda (\bar{d}_j^\lambda C_{\lambda \kappa}^\dagger (1 + \gamma^5) u_j^\kappa) - m_u^\kappa (\bar{d}_j^\lambda C_{\lambda \kappa}^\dagger (1 - \\
 & \gamma^5) u_j^\kappa) - \frac{g}{2} \frac{m_u^\lambda}{M} H (\bar{u}_j^\lambda u_j^\lambda) - \frac{g}{2} \frac{m_d^\lambda}{M} H (\bar{d}_j^\lambda d_j^\lambda) + \frac{i g}{2} \frac{m_u^\lambda}{M} \phi^0 (\bar{u}_j^\lambda \gamma^5 u_j^\lambda) - \\
 & \frac{i g}{2} \frac{m_d^\lambda}{M} \phi^0 (\bar{d}_j^\lambda \gamma^5 d_j^\lambda) + \bar{X}^+ (\partial^2 - M^2) X^+ + \bar{X}^- (\partial^2 - M^2) X^- + \bar{X}^0 (\partial^2 - \\
 & \frac{M^2}{c_w^2}) X^0 + \bar{Y} \partial^2 Y + i g c_w W_\mu^+ (\partial_\mu \bar{X}^0 X^- - \partial_\mu \bar{X}^+ X^0) + i g s_w W_\mu^+ (\partial_\mu \bar{Y} X^- - \\
 & \partial_\mu \bar{X}^+ Y) + i g c_w W_\mu^- (\partial_\mu \bar{X}^- X^0 - \partial_\mu \bar{X}^0 X^+) + i g s_w W_\mu^- (\partial_\mu \bar{X}^- Y - \\
 & \partial_\mu \bar{Y} X^+) + i g c_w Z_\mu^0 (\partial_\mu \bar{X}^+ X^+ - \partial_\mu \bar{X}^- X^-) + i g s_w A_\mu (\partial_\mu \bar{X}^+ X^+ - \\
 & \partial_\mu \bar{X}^- X^-) - \frac{1}{2} g M [\bar{X}^+ X^+ H + \bar{X}^- X^- H + \frac{1}{c_w^2} \bar{X}^0 X^0 H] + \\
 & \frac{1-2c_w^2}{2c_w} i g M [\bar{X}^+ X^0 \phi^+ - \bar{X}^- X^0 \phi^-] + \frac{1}{2 c_w} i g M [\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \\
 & i g M s_w [\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \frac{1}{2} i g M [\bar{X}^+ X^+ \phi^0 - \bar{X}^- X^- \phi^0]
 \end{aligned}$$

credit: symmetry magazine

Higgs Production at LEP (1989-2000)



$H^0 \rightarrow b\bar{b}$ dominant (BR $\approx 84\%$)

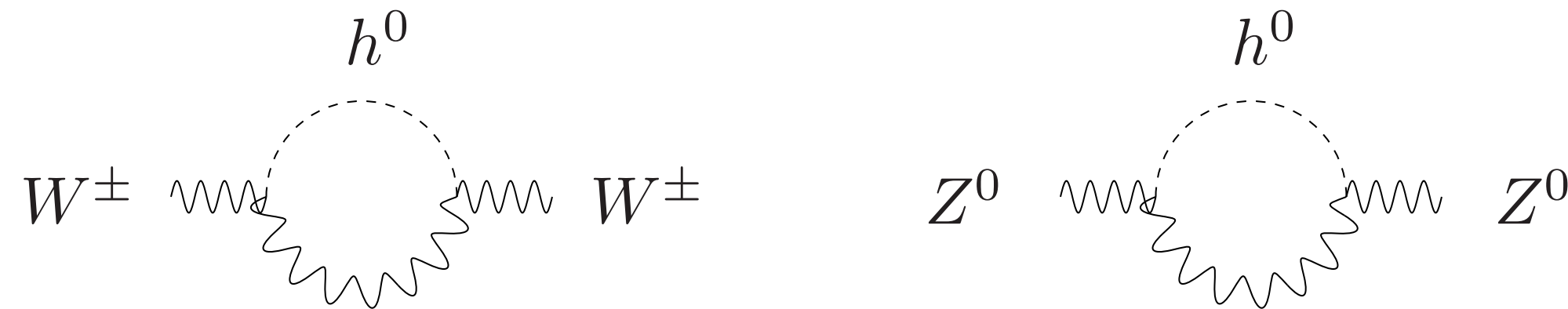
$H^0(\rightarrow b\bar{b})Z(\rightarrow q\bar{q}) \sim 60\%$

$H^0(\rightarrow b\bar{b})Z(\rightarrow \nu\bar{\nu}) \sim 17\%$

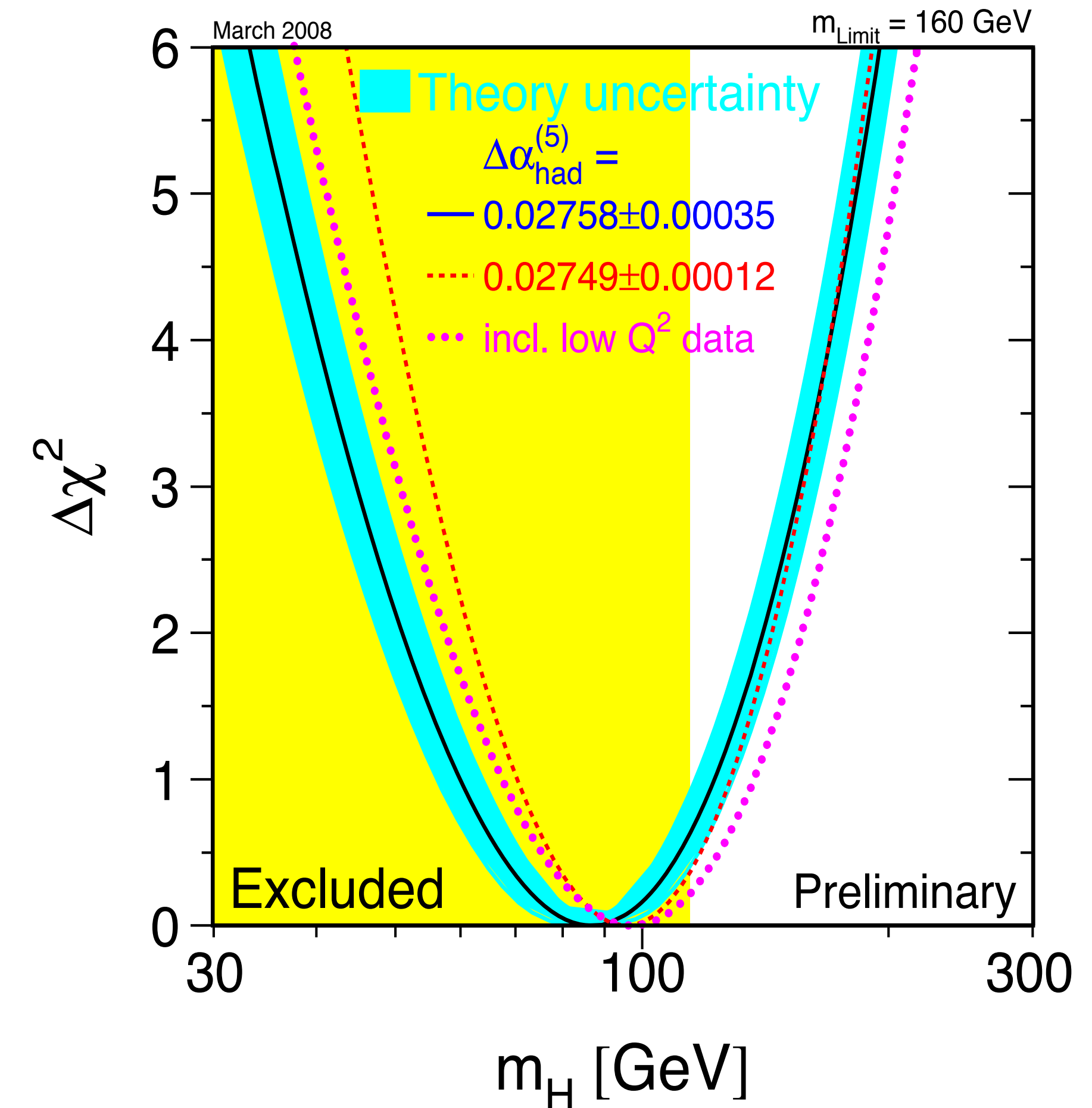
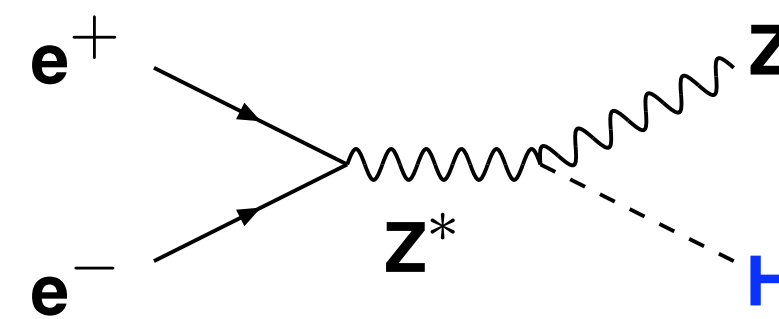
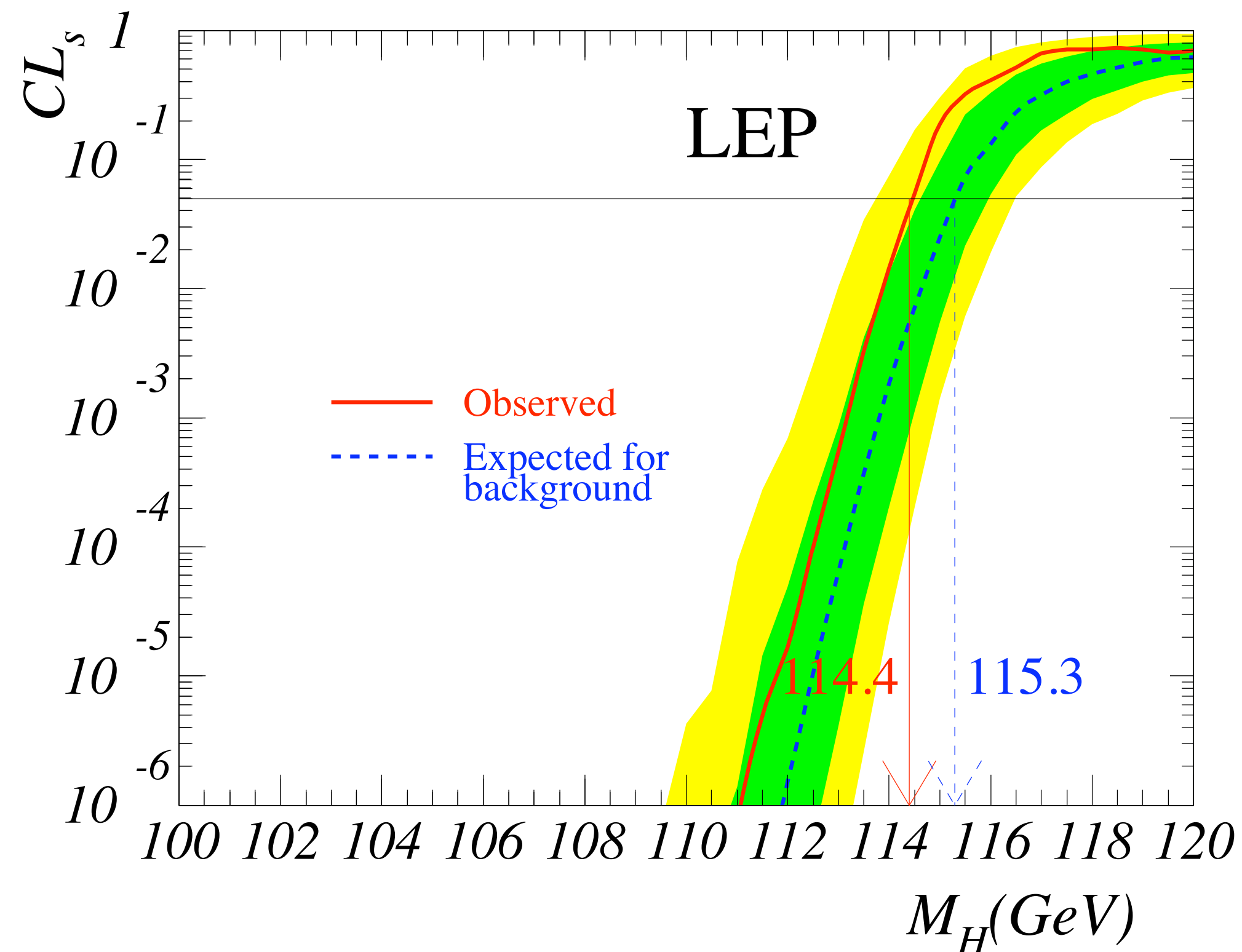
$H^0(\rightarrow b\bar{b})Z(\rightarrow \ell^+\ell^-)$ and $H^0(\rightarrow \tau^+\tau^-)Z(\rightarrow q\bar{q}) \sim 14\%$

Experimental Constraints on Higgs Mass @ LEP (~2000)

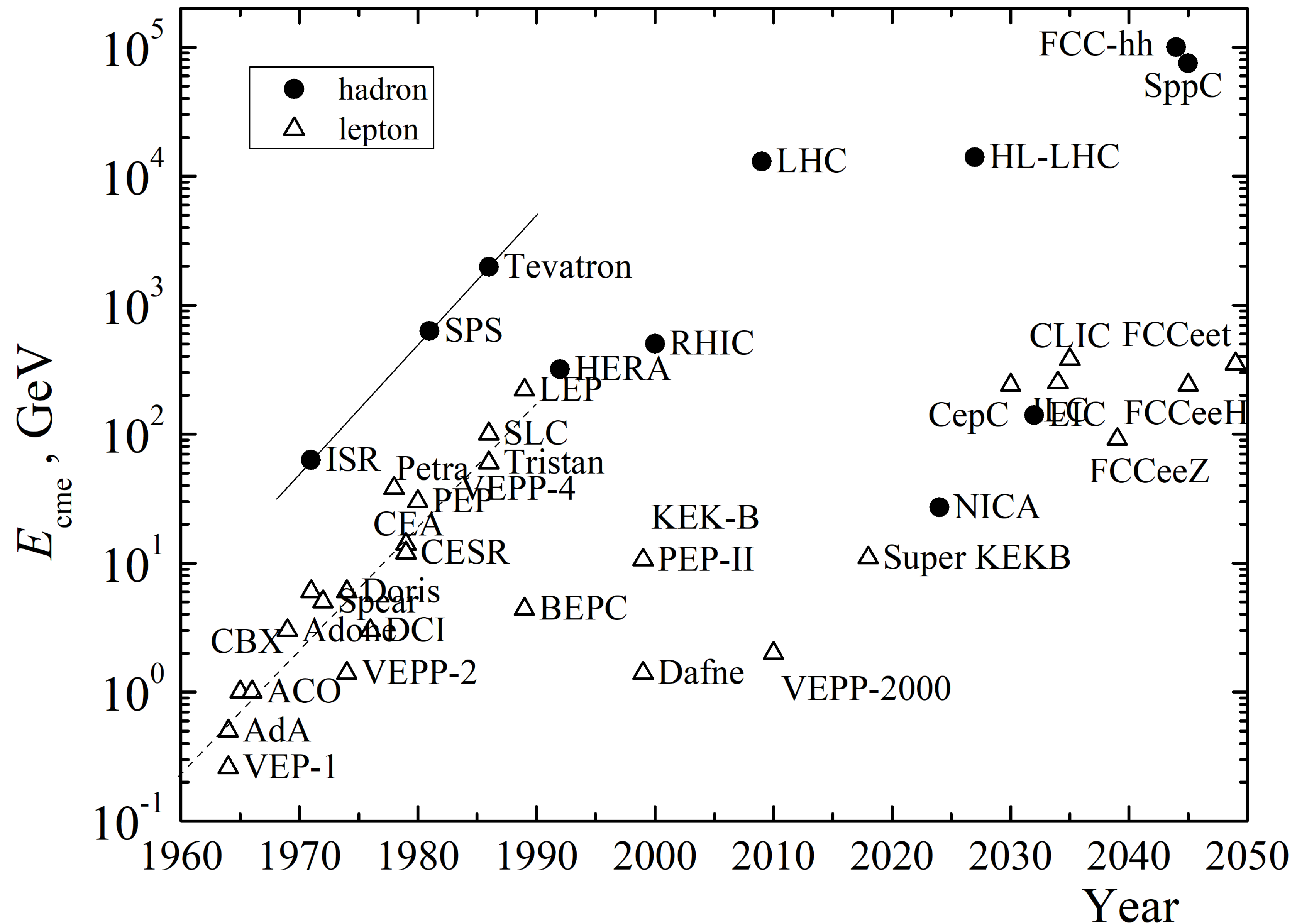
- ▷ Indirect search through precision measurement of SM sensitive to radiative corrections to Higgs in loops



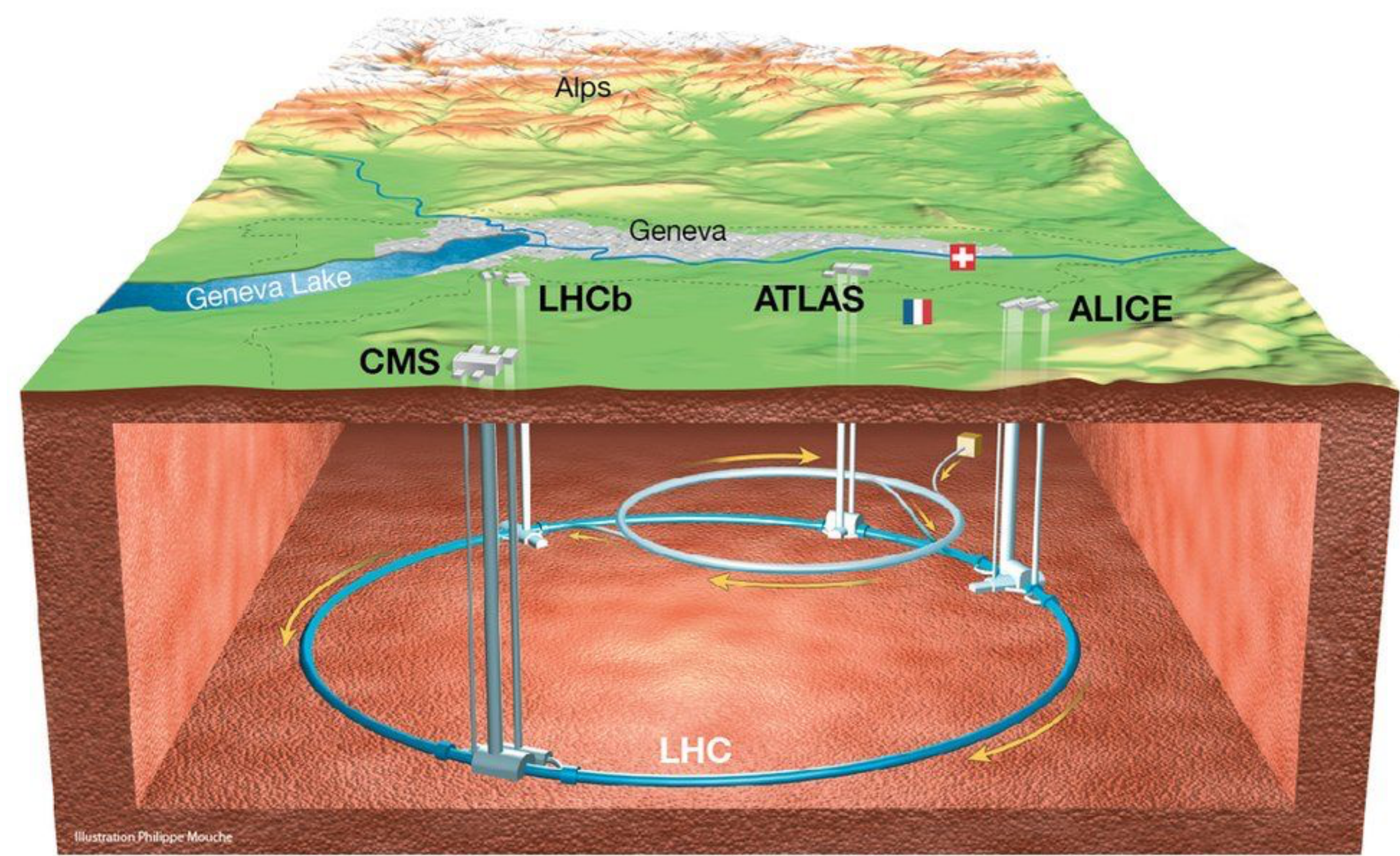
- ▷ Direct search at LEP in Higgs-strahlung production channel



Evolution of Particle Colliders



Large Hadron Collider at CERN



Year(s)	Run	Center-of-mass energy $\sqrt{s}$
2009	Commissioning	0.9 TeV and 2.36 TeV
2010	Run 1	7 TeV
2011	Run 1	7 TeV
2012	Run 1	8 TeV
2013–2014	Long Shutdown 1	—
2015	Run 2	13 TeV
2016	Run 2	13 TeV
2017	Run 2	13 TeV
2018	Run 2	13 TeV
2019–2021	Long Shutdown 2	—
2022	Run 3	13.6 TeV
2023	Run 3	13.6 TeV
2024	Run 3	13.6 TeV
2025	Run 3	13.6 TeV
2026	Run 3	13.6 TeV

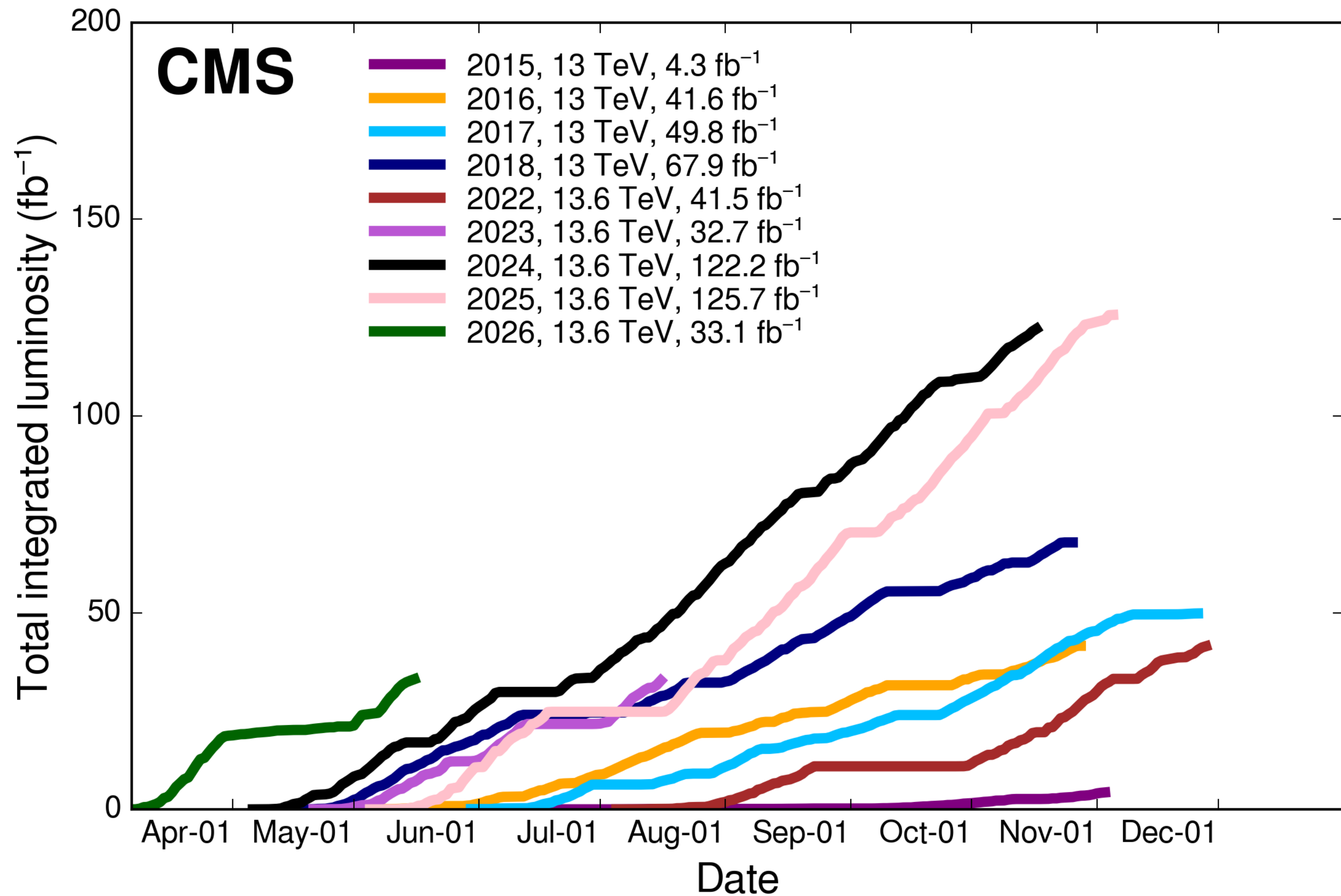
Design : $\sqrt{s} = 14 \text{ TeV}$

Will be reached in 2031!

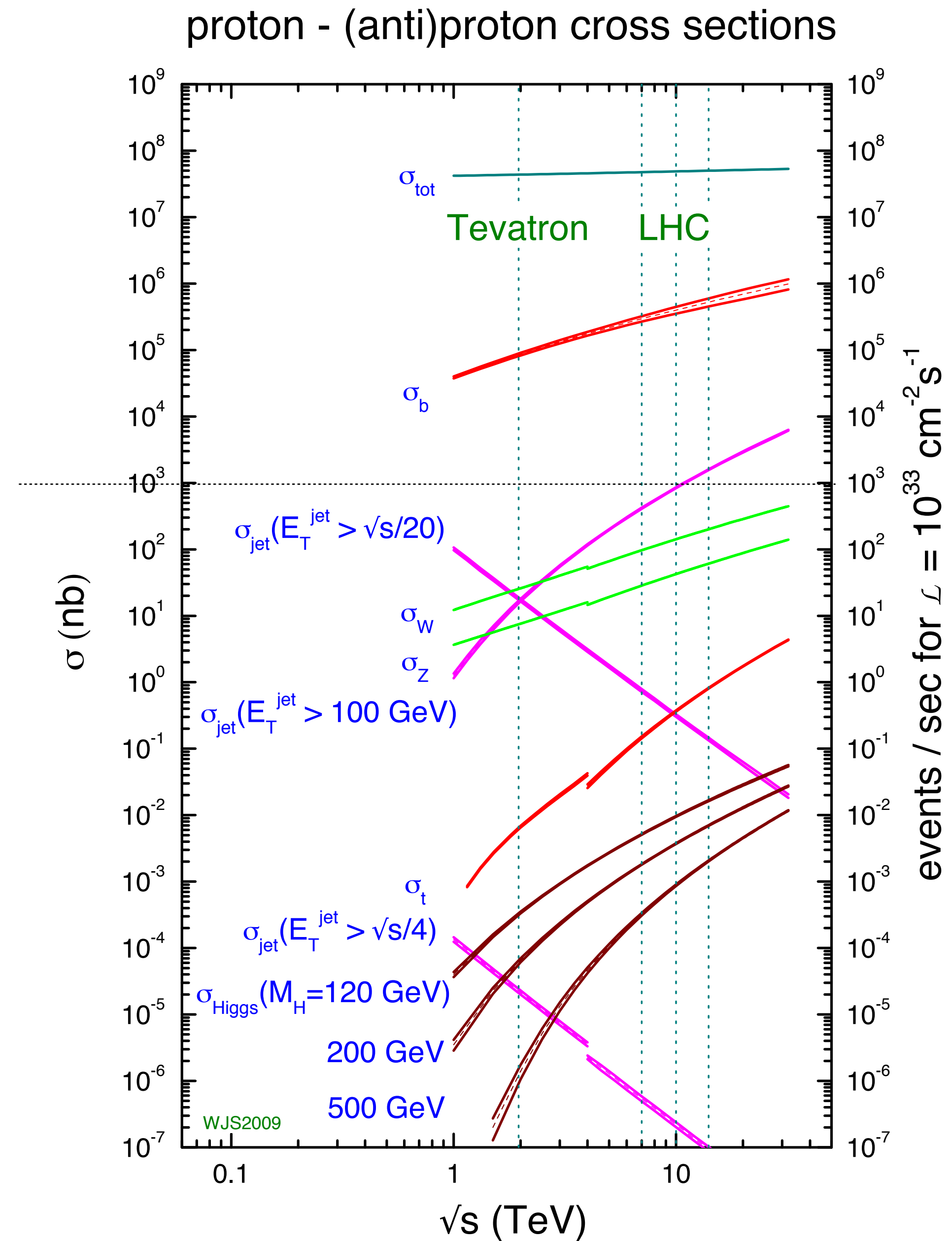
Four Experiments at LHC



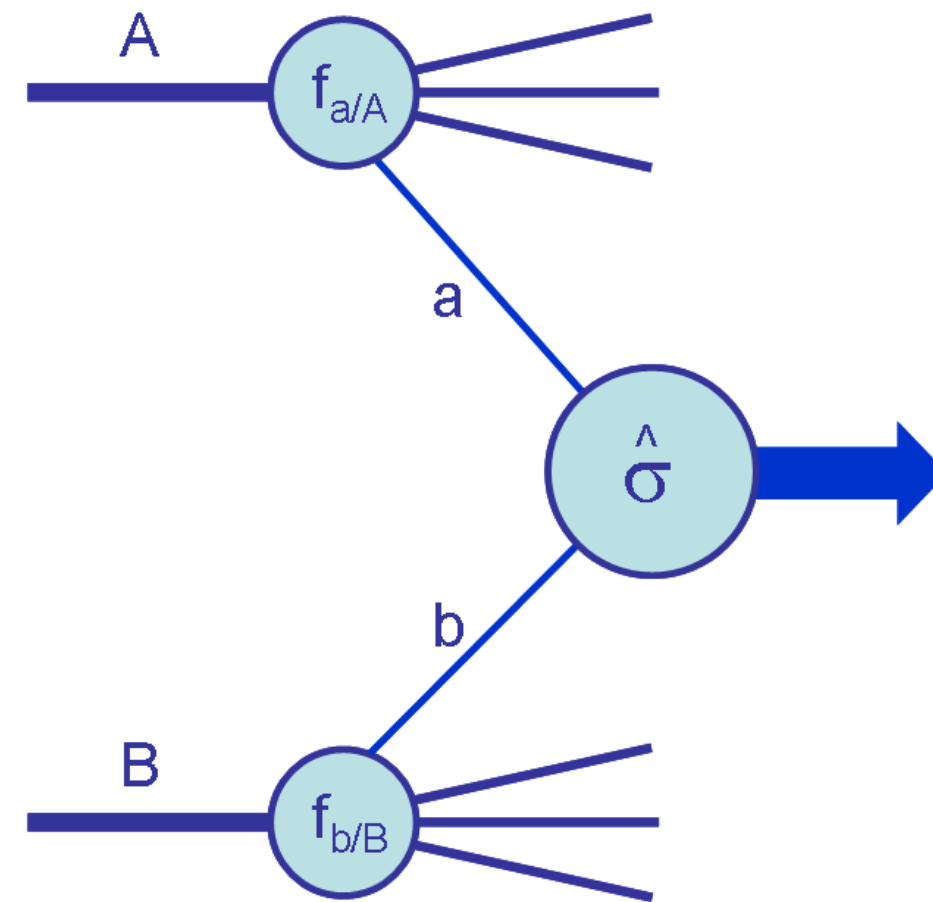
LHC Integrated Luminosity



Physics Production at LHC



Proton-proton Cross Section



$$\sigma_{AB} = \sum_{a,b=q,g} \left[\hat{\sigma}_{ab}^{\text{LO}} + \alpha_S(Q^2) \hat{\sigma}_{ab}^{\text{NLO}} + \dots \right] \otimes f_{a/A}(x_a, Q^2) \otimes f_{b/B}(x_b, Q^2)$$

$$\sigma_X = \sum_{a,b} \int_0^1 dx_a dx_b f(x_a, \text{flav}_a, Q^2) f(x_b, \text{flav}_b, Q^2) \cdot \sigma_{ab \rightarrow X}(x_a, x_b, Q^2)$$

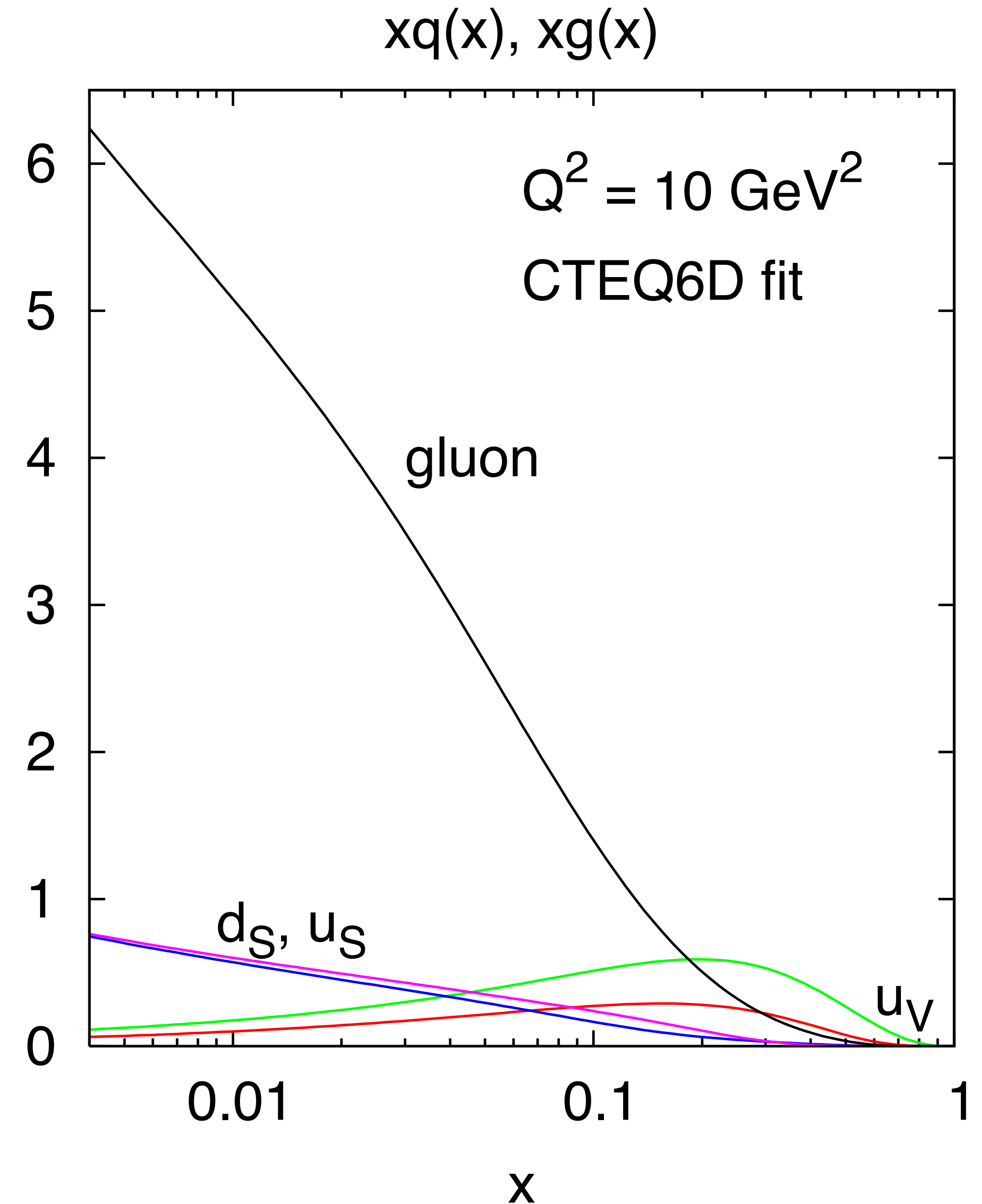
Sum over initial partonic states a,b

Parton Density Function

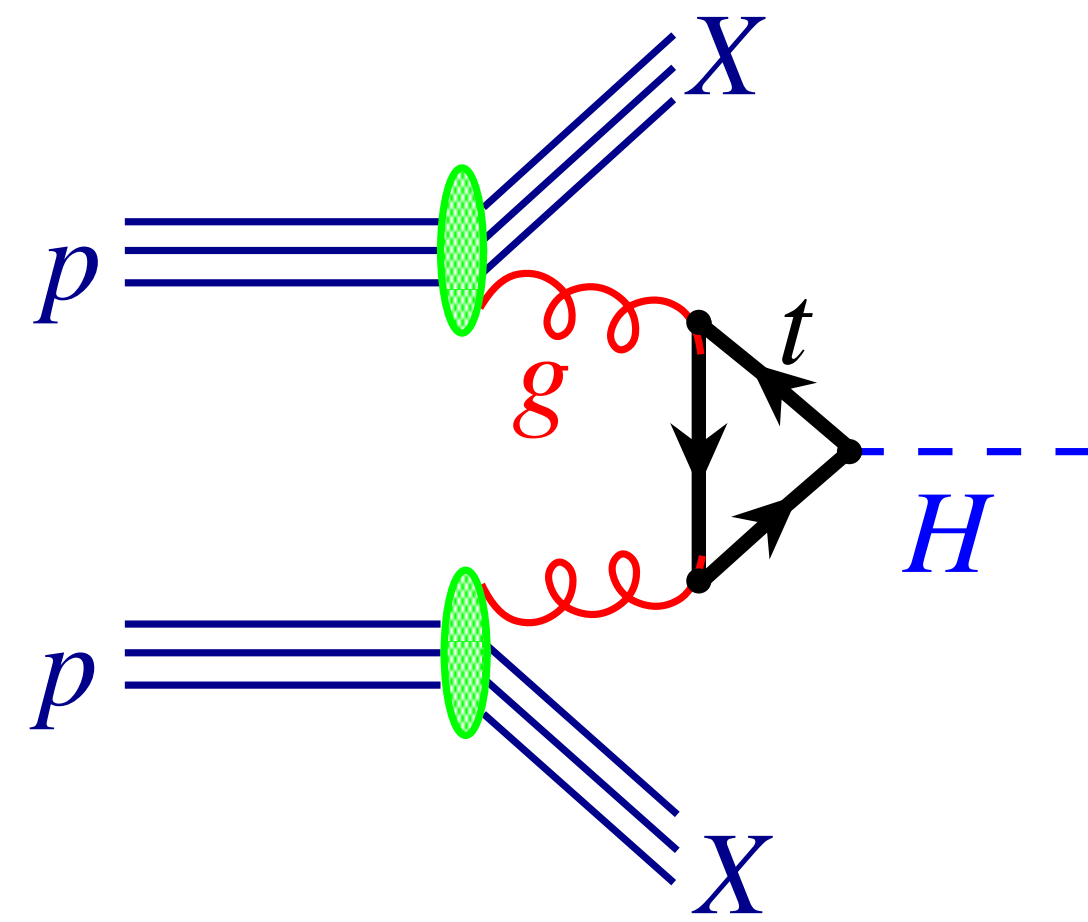
hard scattering cross-section

Gluon Density Function

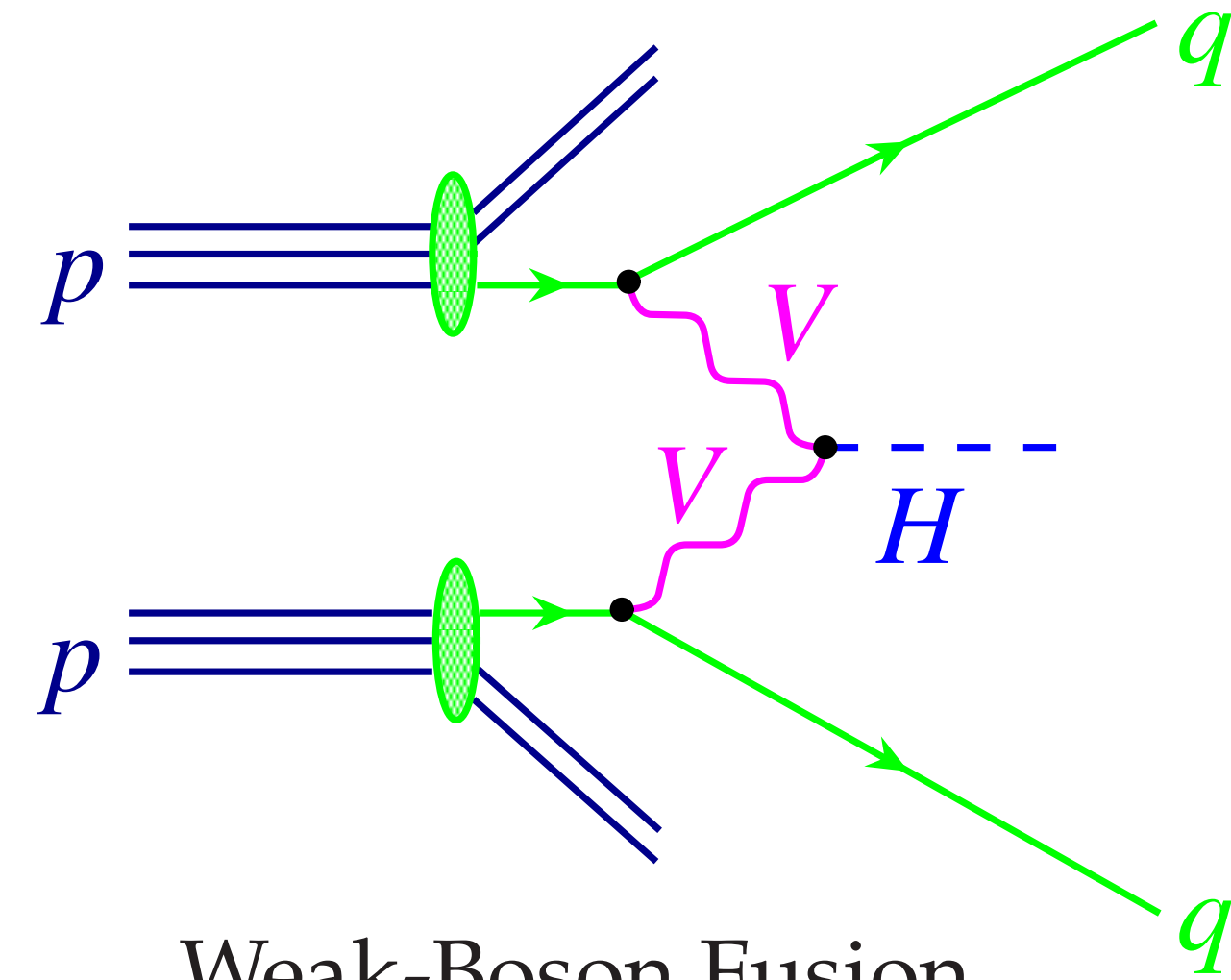
- ▷ Gluons dominate by far at low x
- ▷ LHC dominated mainly by gluon-gluon fusion (hard scattering of 2 gluons) at low x
- ▷ Different experimental signatures for qq , q -anti- q and qg hard scattering
 - also very different cross sections



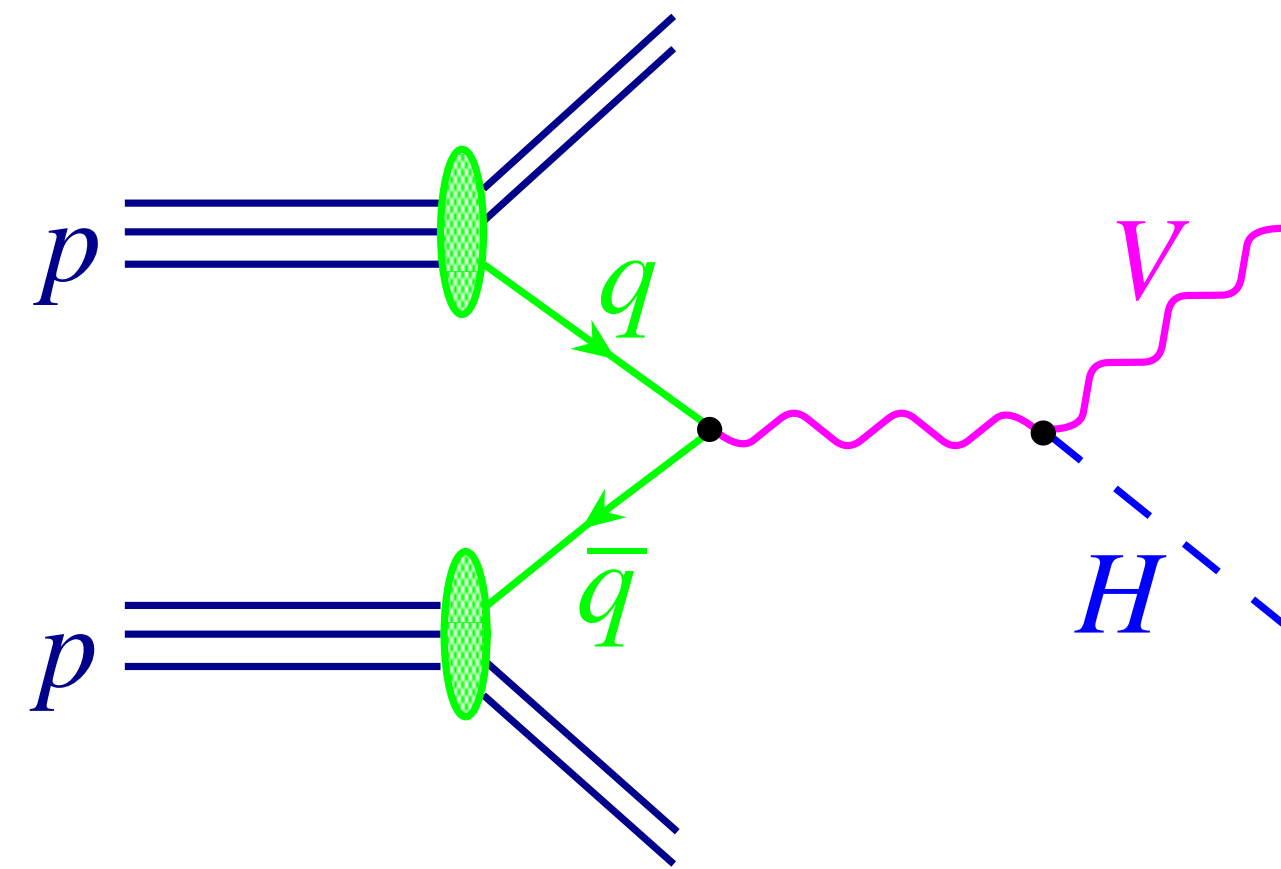
Higgs Production at Hadron Colliders



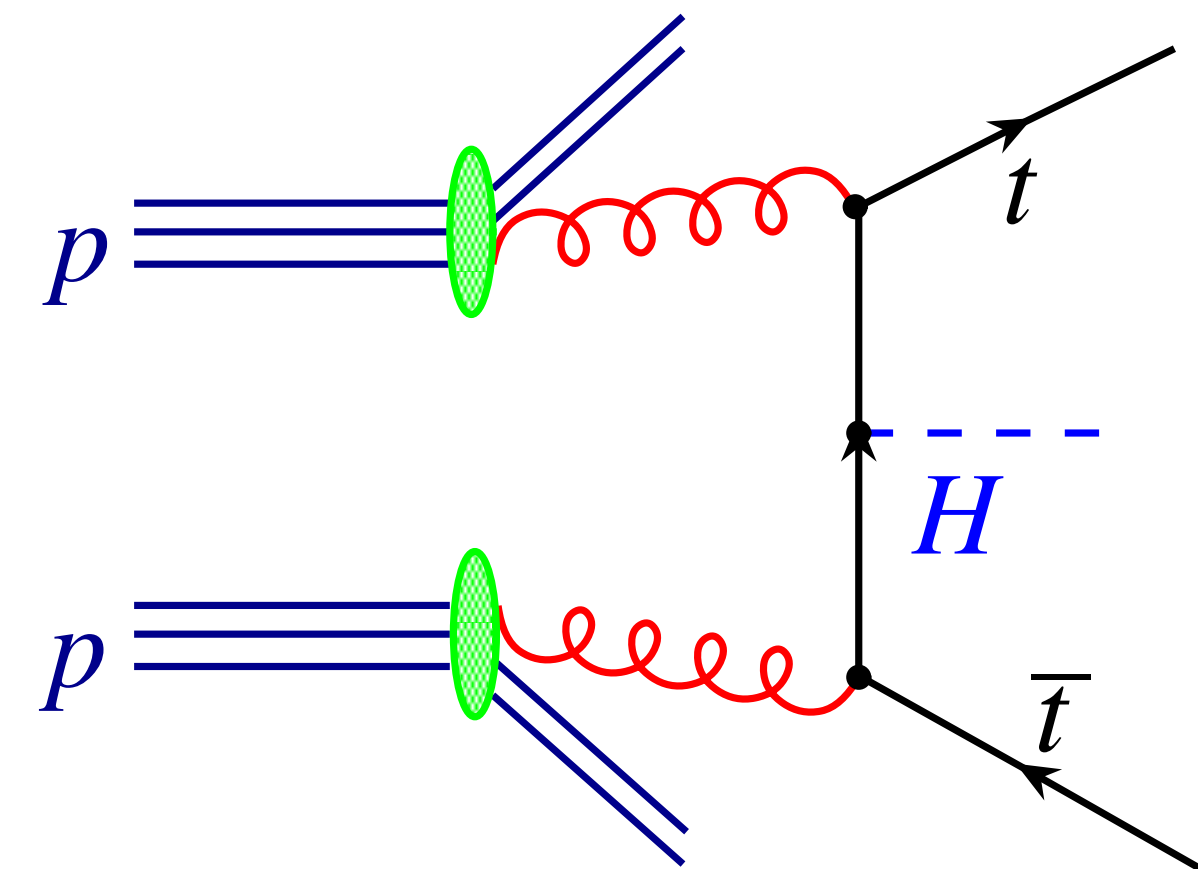
Gluon fusion



Weak-Boson Fusion

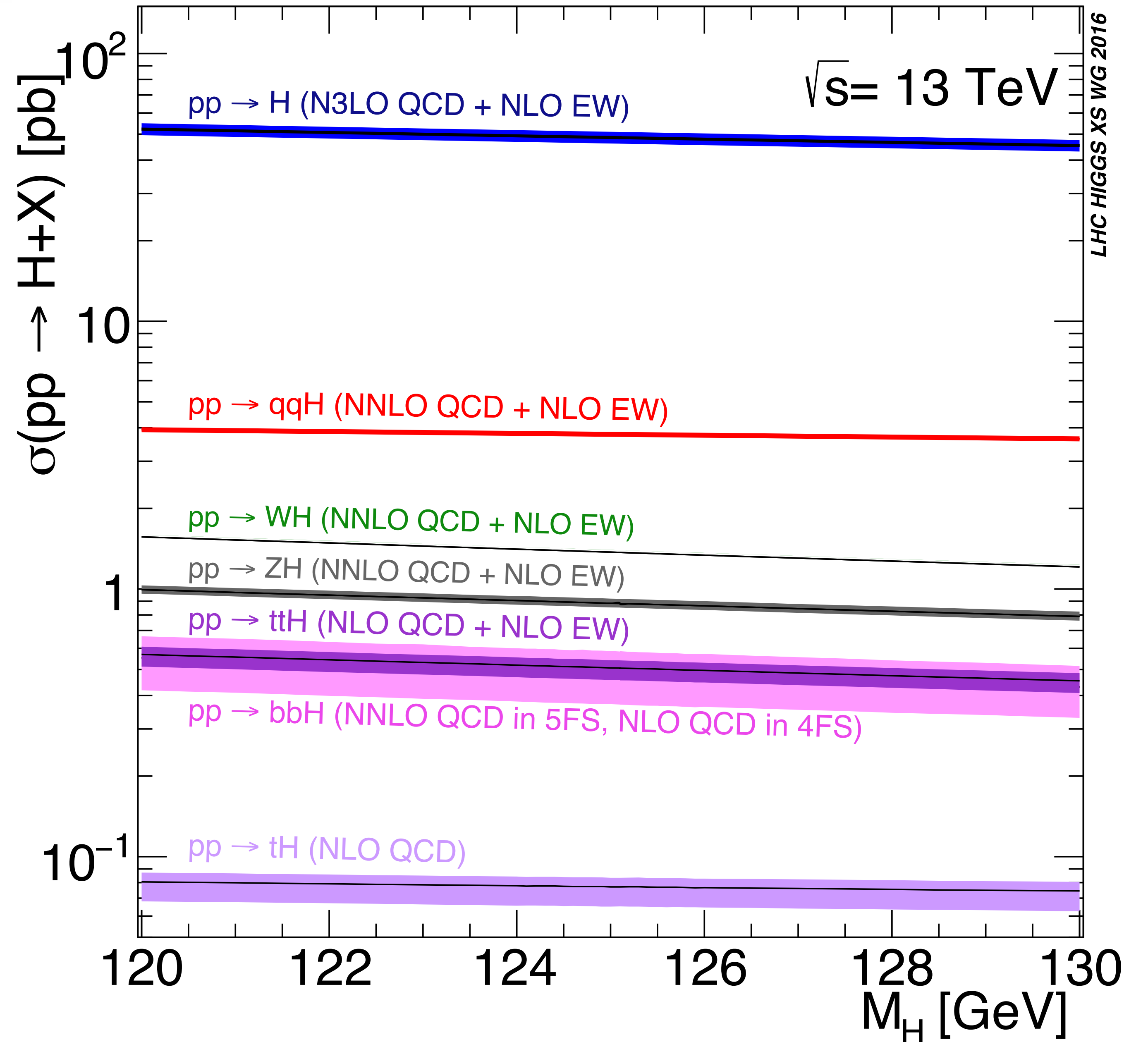
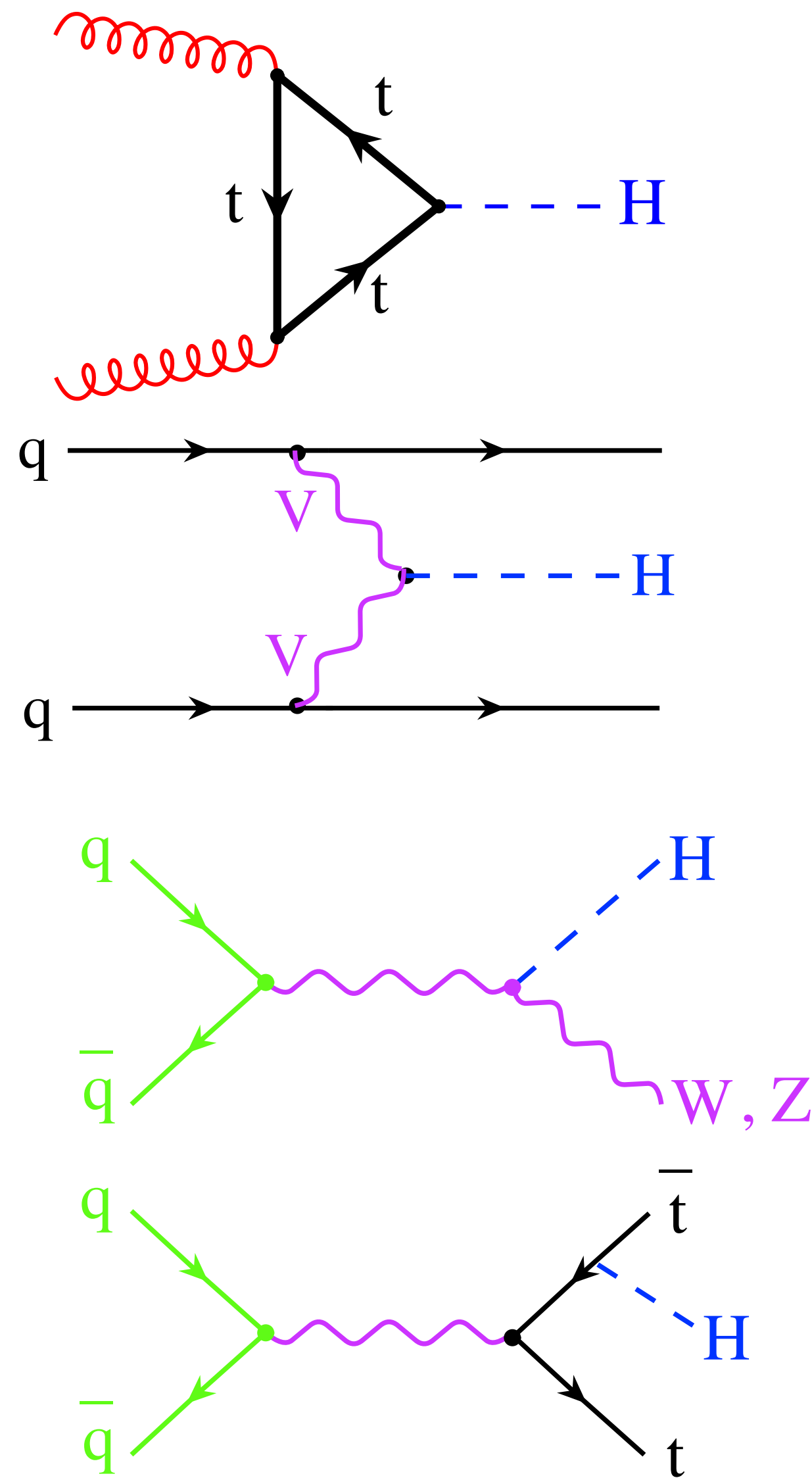


Higgs Strahlung

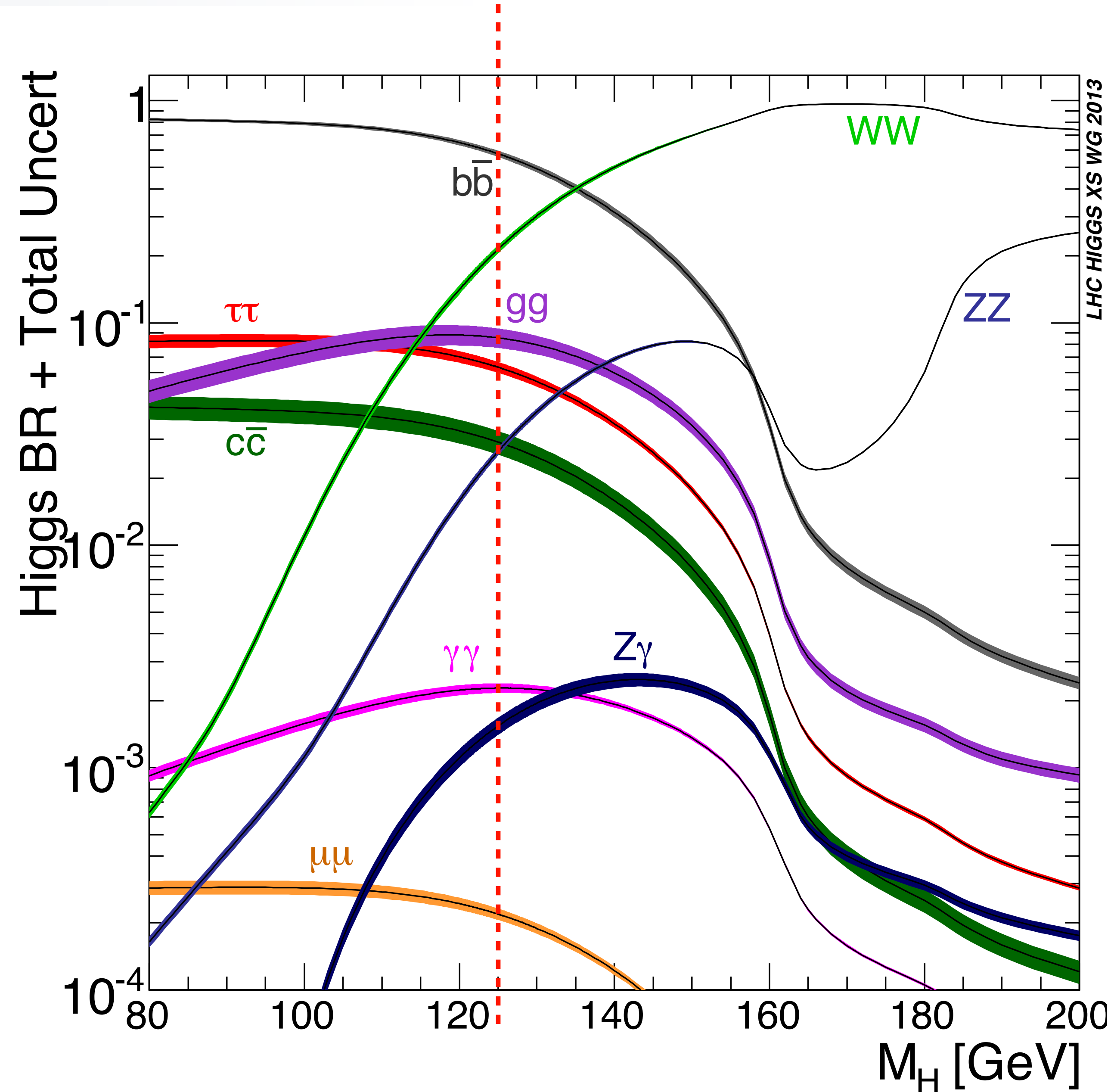


$t\bar{t}H$

Higgs Production Cross Section

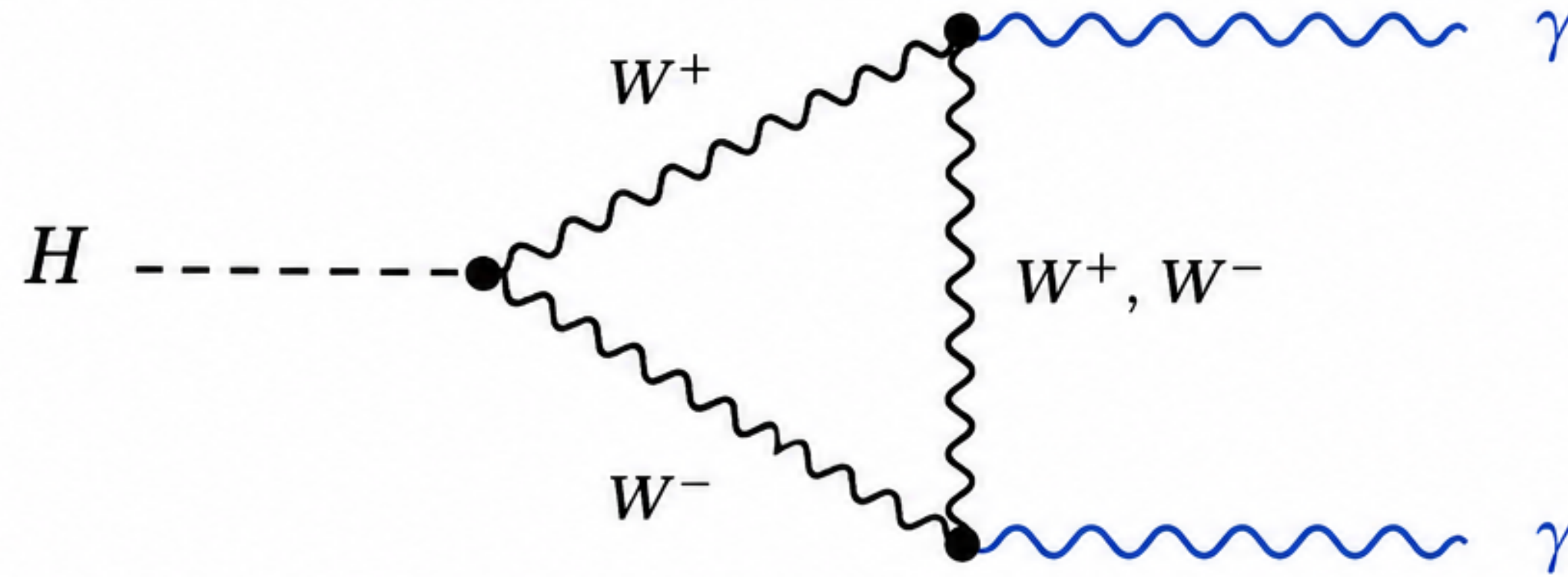


Higgs branching fractions

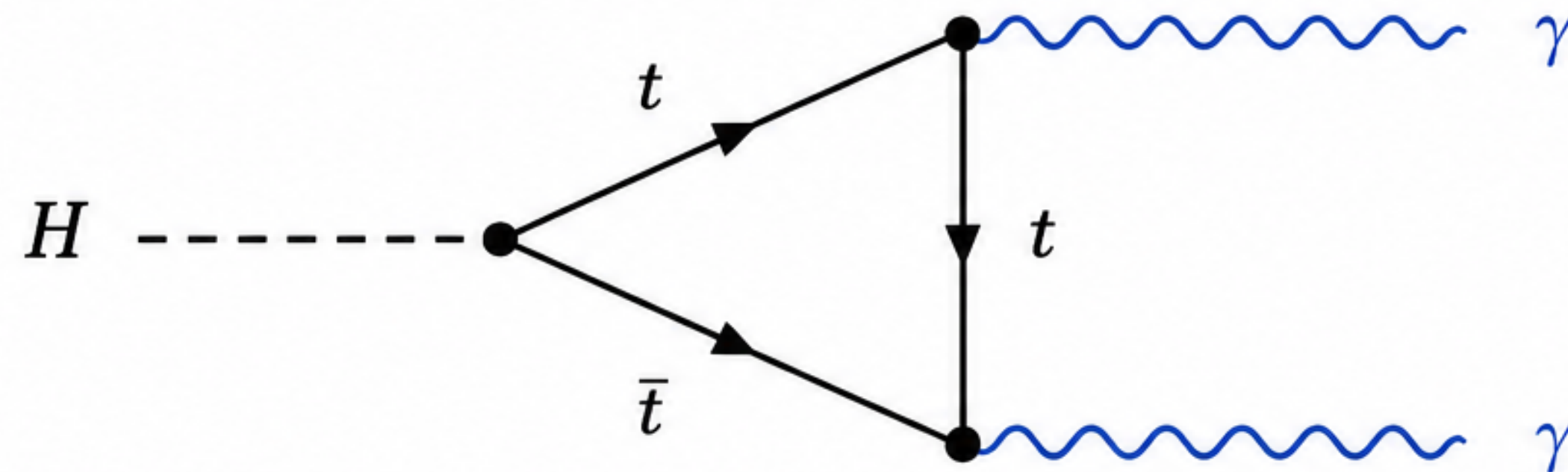


Higgs decay to photons

(a) W boson loop (dominant)



(b) Top quark loop (subdominant)



- H has no tree-level coupling to photons.
- Decay proceeds via loops of charged particles.
- Main contributions:
 - (a) W boson loop
 - (b) Top quark loop

The W loop interferes destructively with the top loop.

The total decay is dominated by the W loop.

Decay width (at leading order):

$$\Gamma(H \rightarrow \gamma\gamma) = \frac{G_F \alpha^2 m_H^3}{128 \sqrt{2} \pi^3} \left| A_W(\tau_W) + \sum_f N_c Q_f^2 A_f(\tau_f) \right|^2$$

$$\tau_i = \frac{4m_i^2}{m_H^2} \quad \begin{array}{l} N_c: \text{color factor} \\ Q_f: \text{electric charge of fermion } f \end{array}$$

Candidate Higgs Events



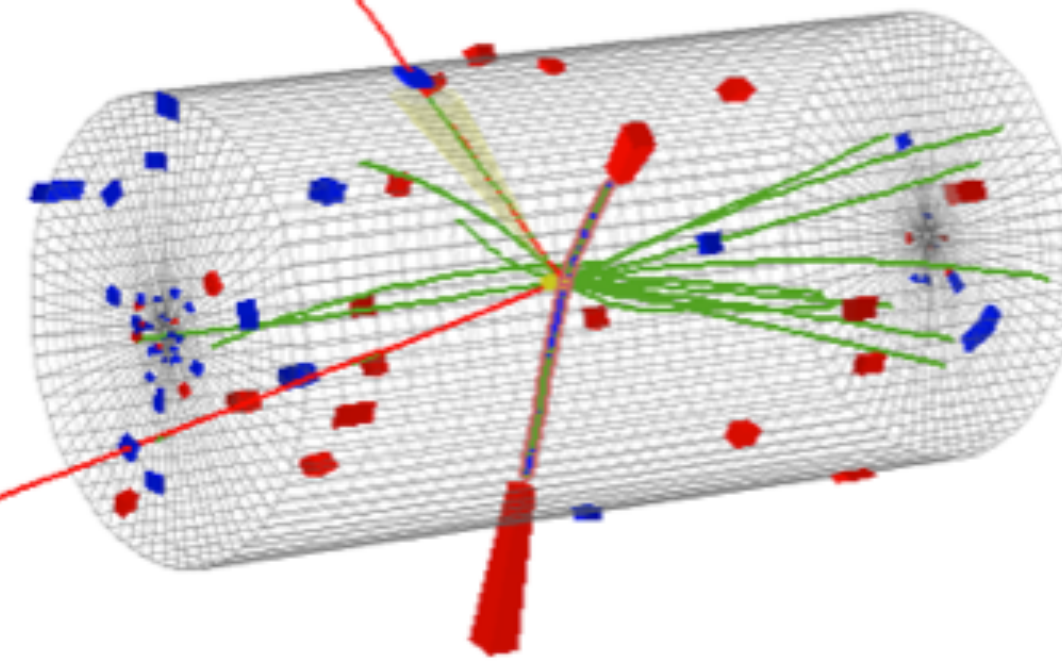
$\mu^+(Z_1) p_T : 43 \text{ GeV}$

8 TeV DATA

4-lepton Mass : 126.9 GeV

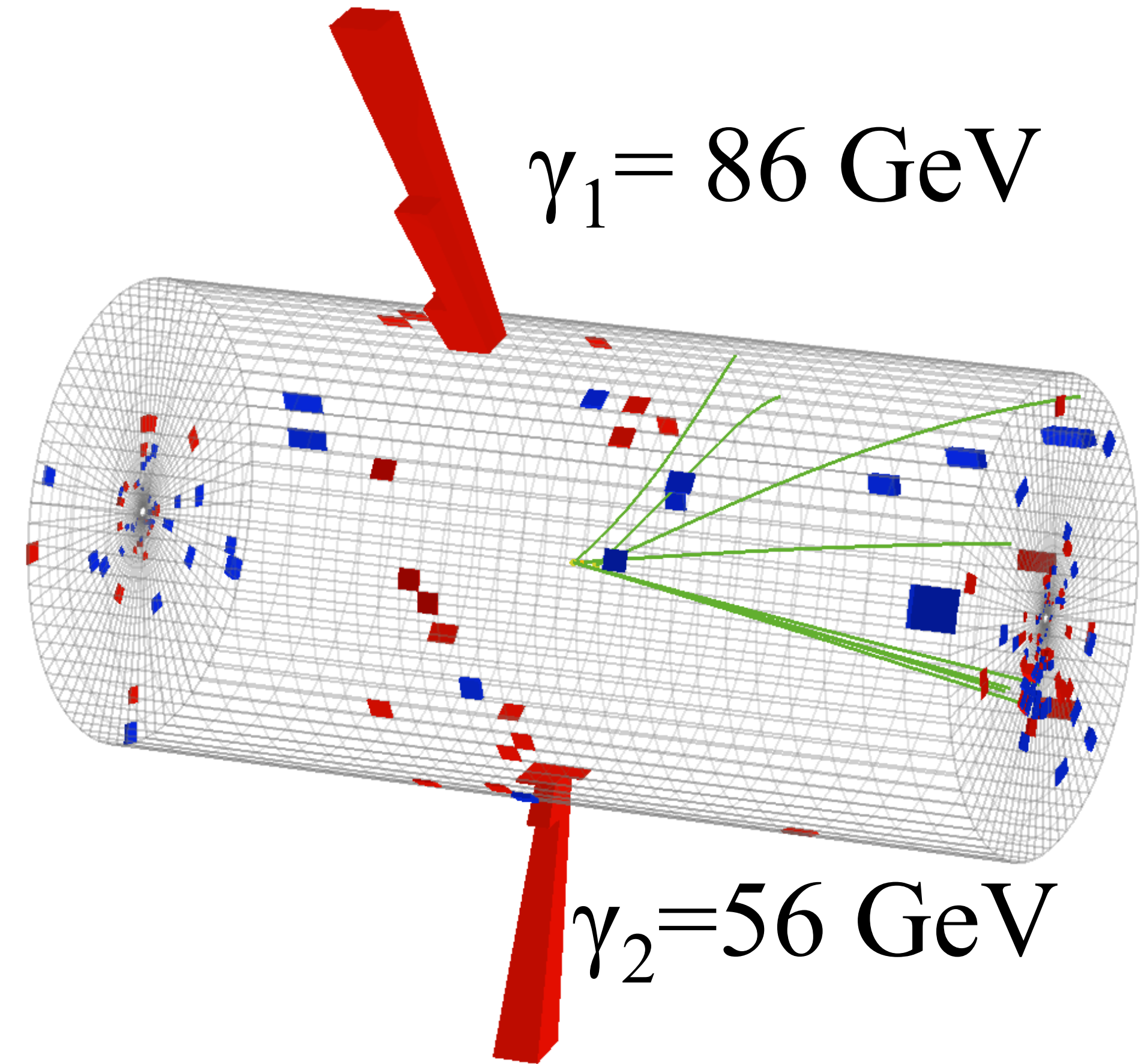
$\mu^-(Z_1) p_T : 24 \text{ GeV}$

$e^-(Z_2) p_T : 10 \text{ GeV}$

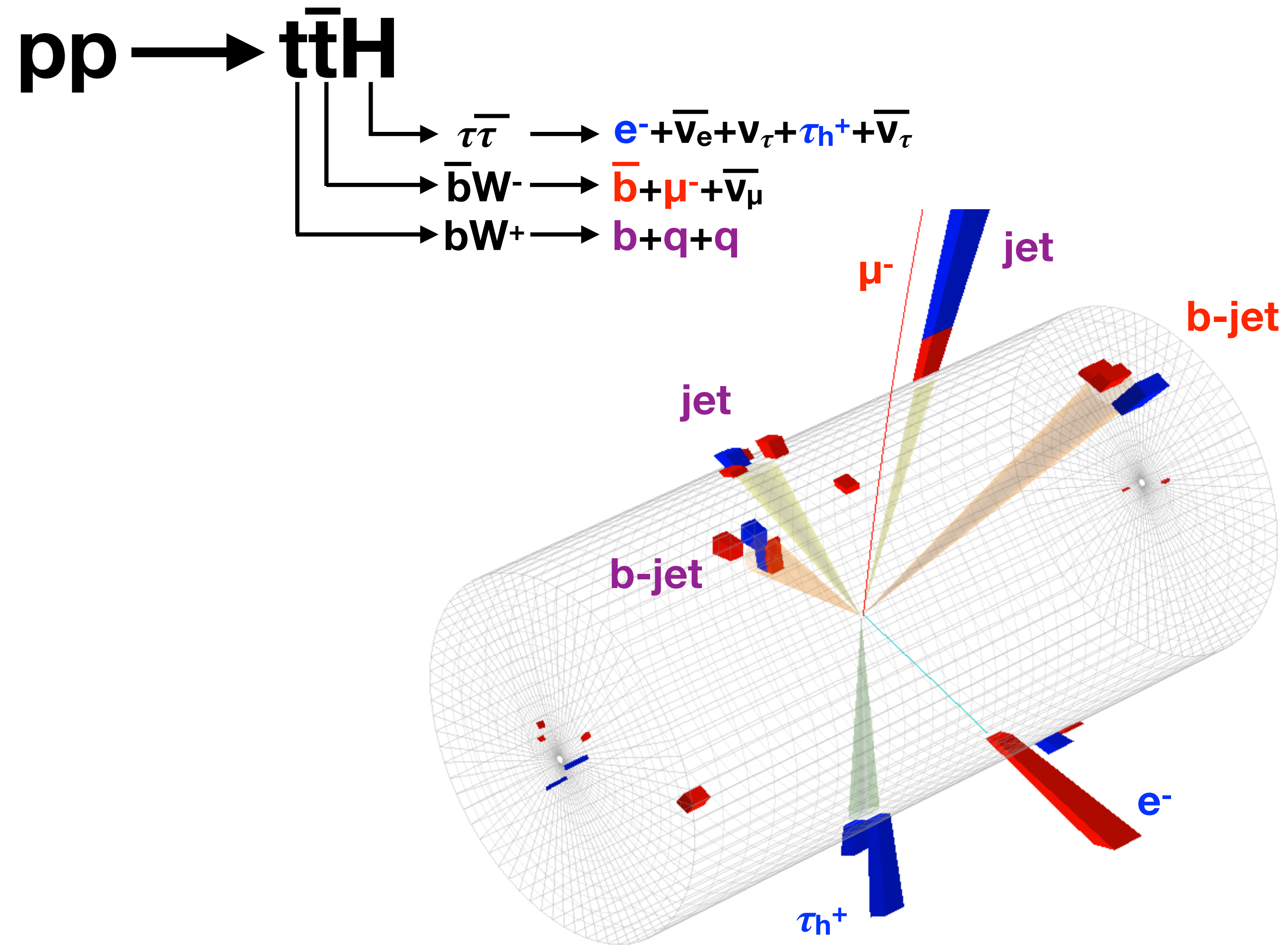


$e^+(Z_2) p_T : 21 \text{ GeV}$

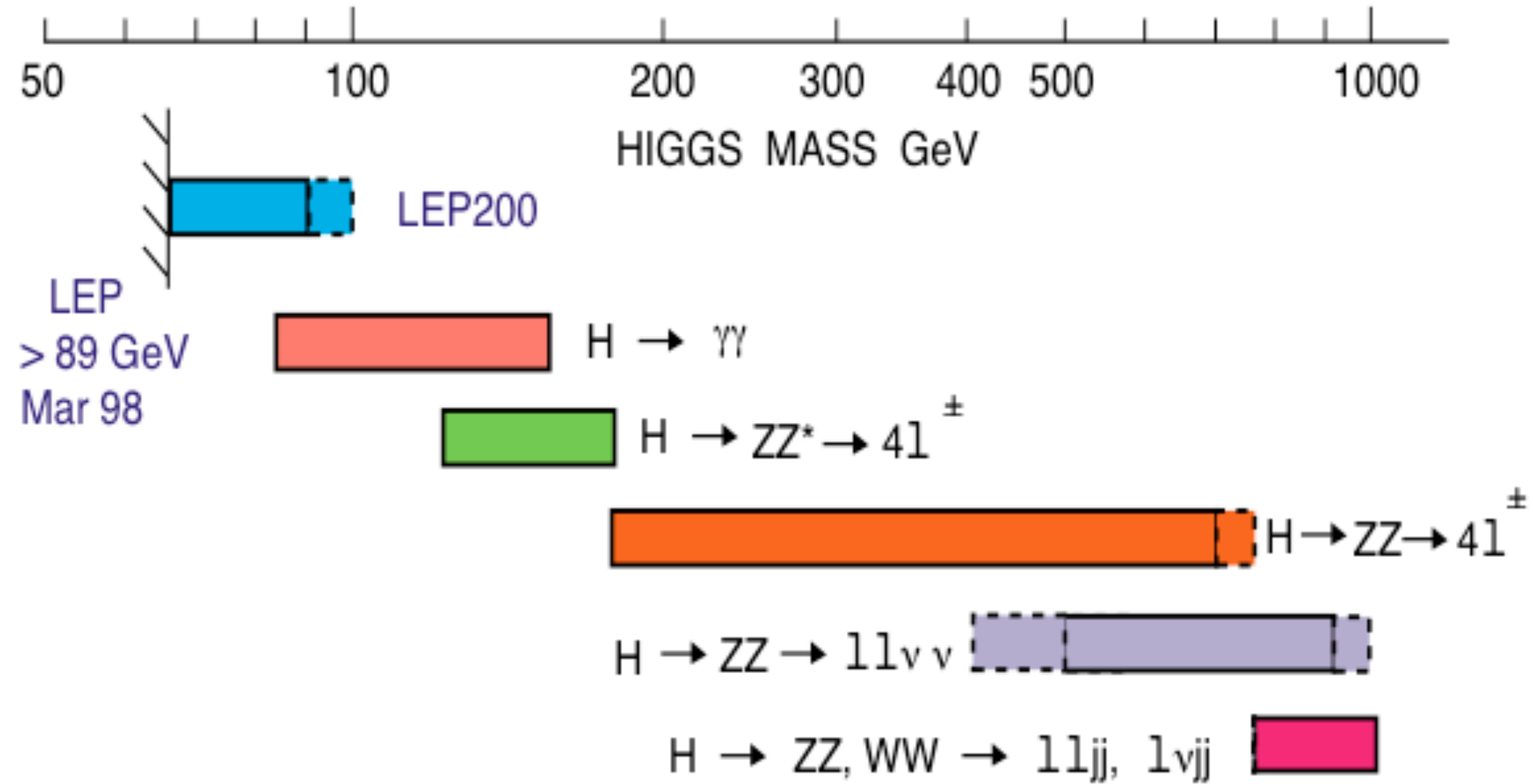
CMS Experiment at LHC, CERN
Data recorded: Mon May 28 01:35:47 2012 CEST
Run/Event: 195099 / 137440354
Lumi section: 115



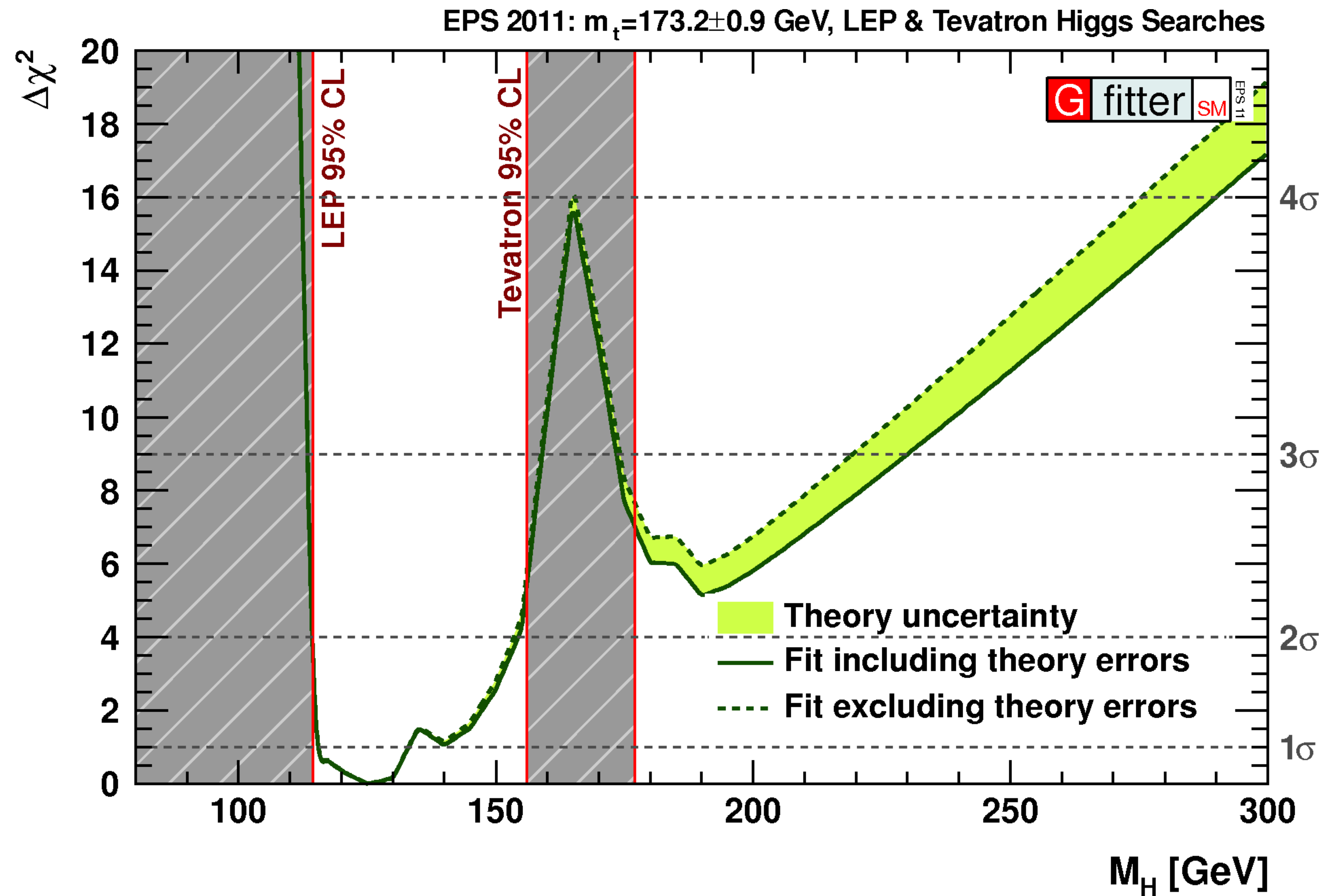
Rare Higgs event



Higgs Discovery Channels

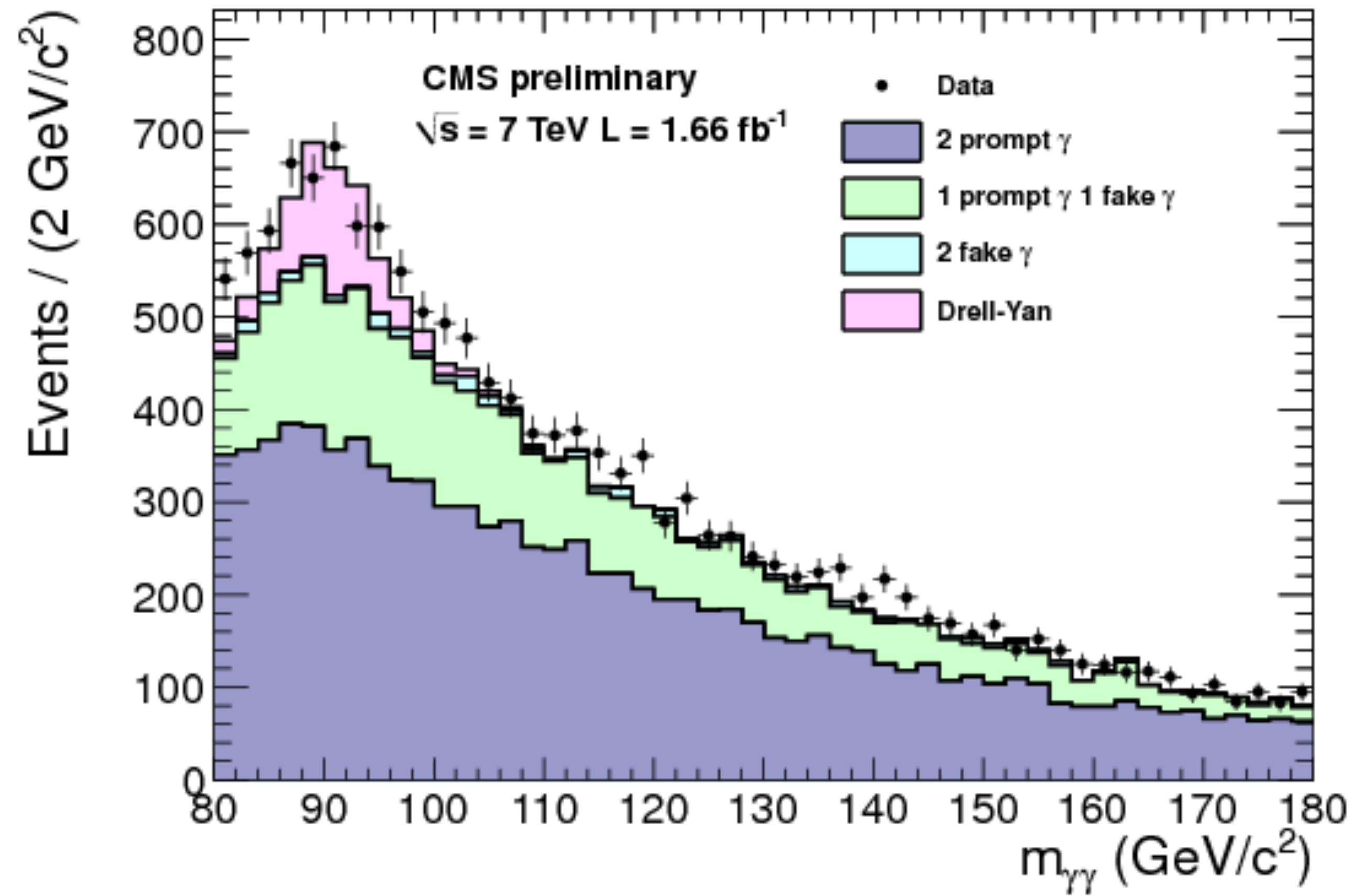
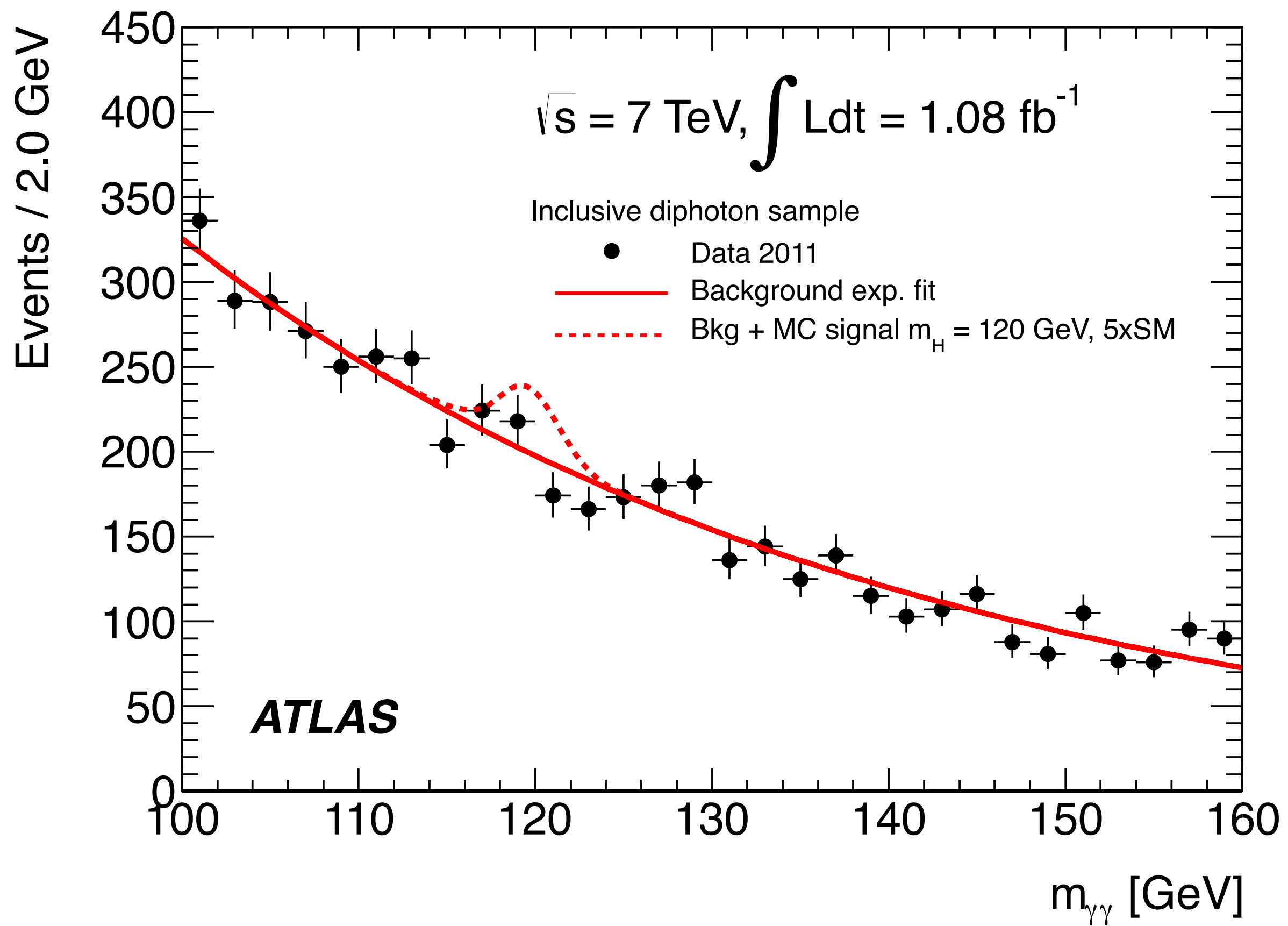


Higgs Searches before LHC @ Summer 2011

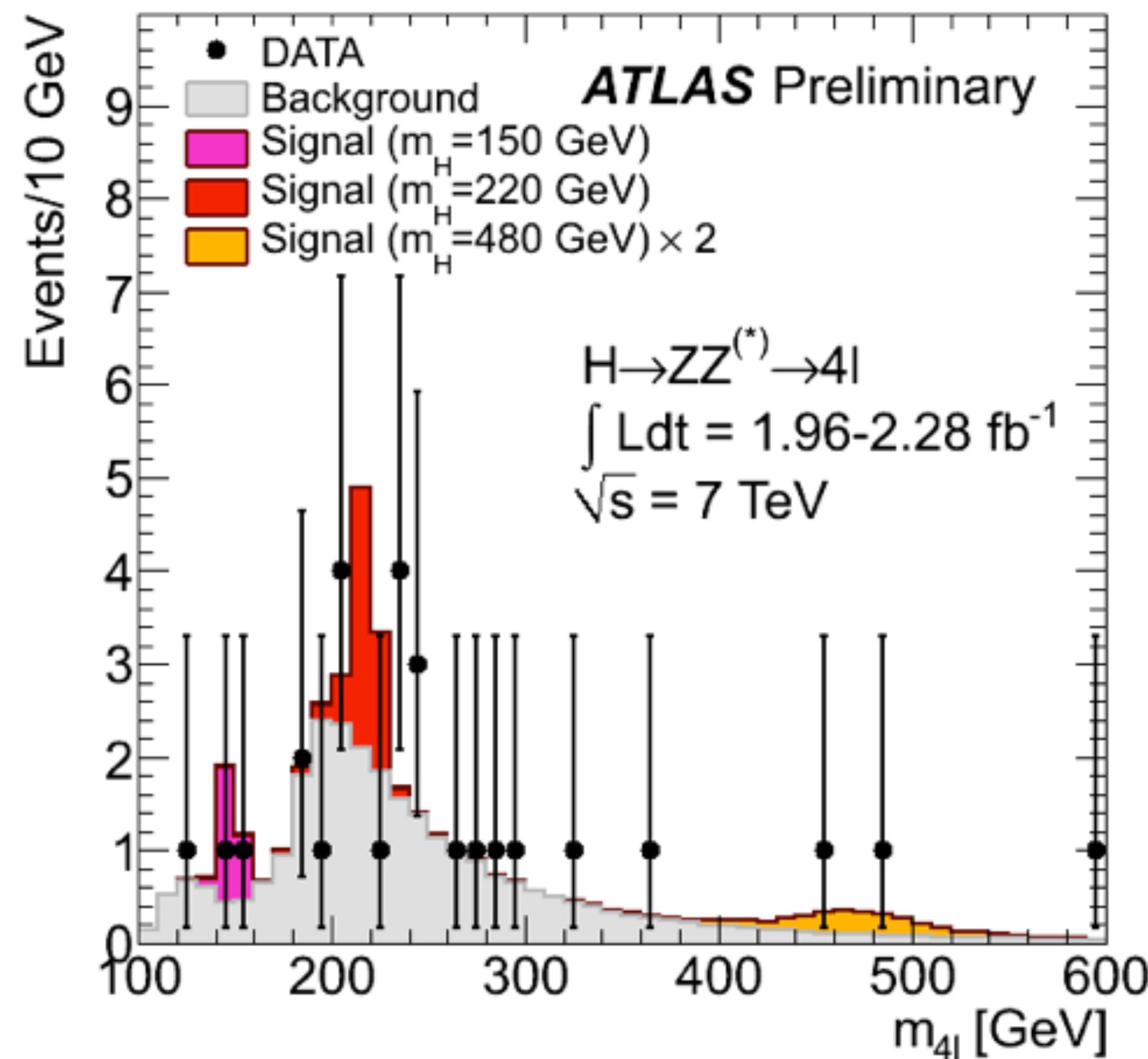


<http://arxiv.org/abs/1107.0975>

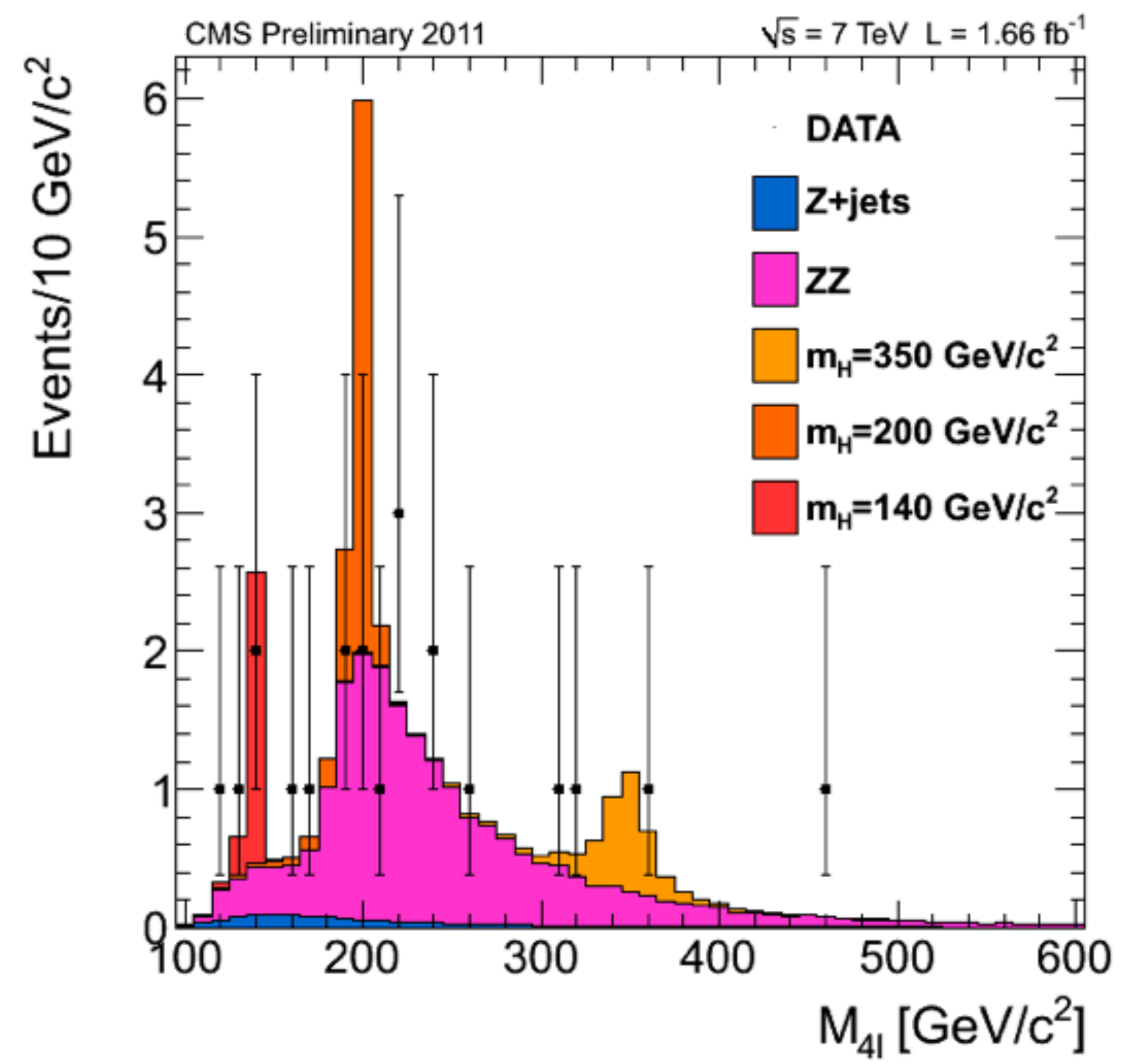
H $\rightarrow\gamma\gamma$ Early 2011



$H \rightarrow ZZ \rightarrow 4l$ Early 2011

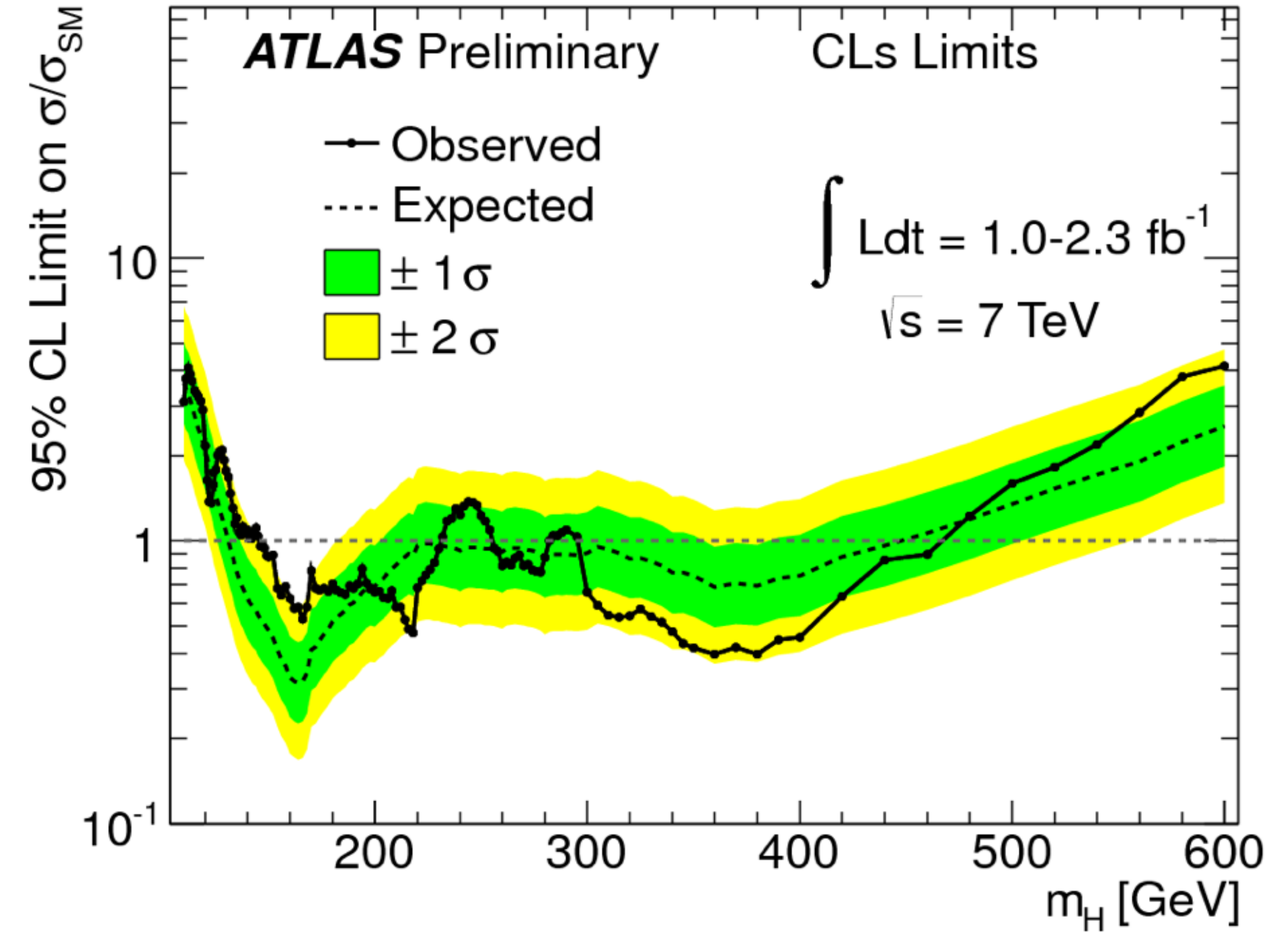
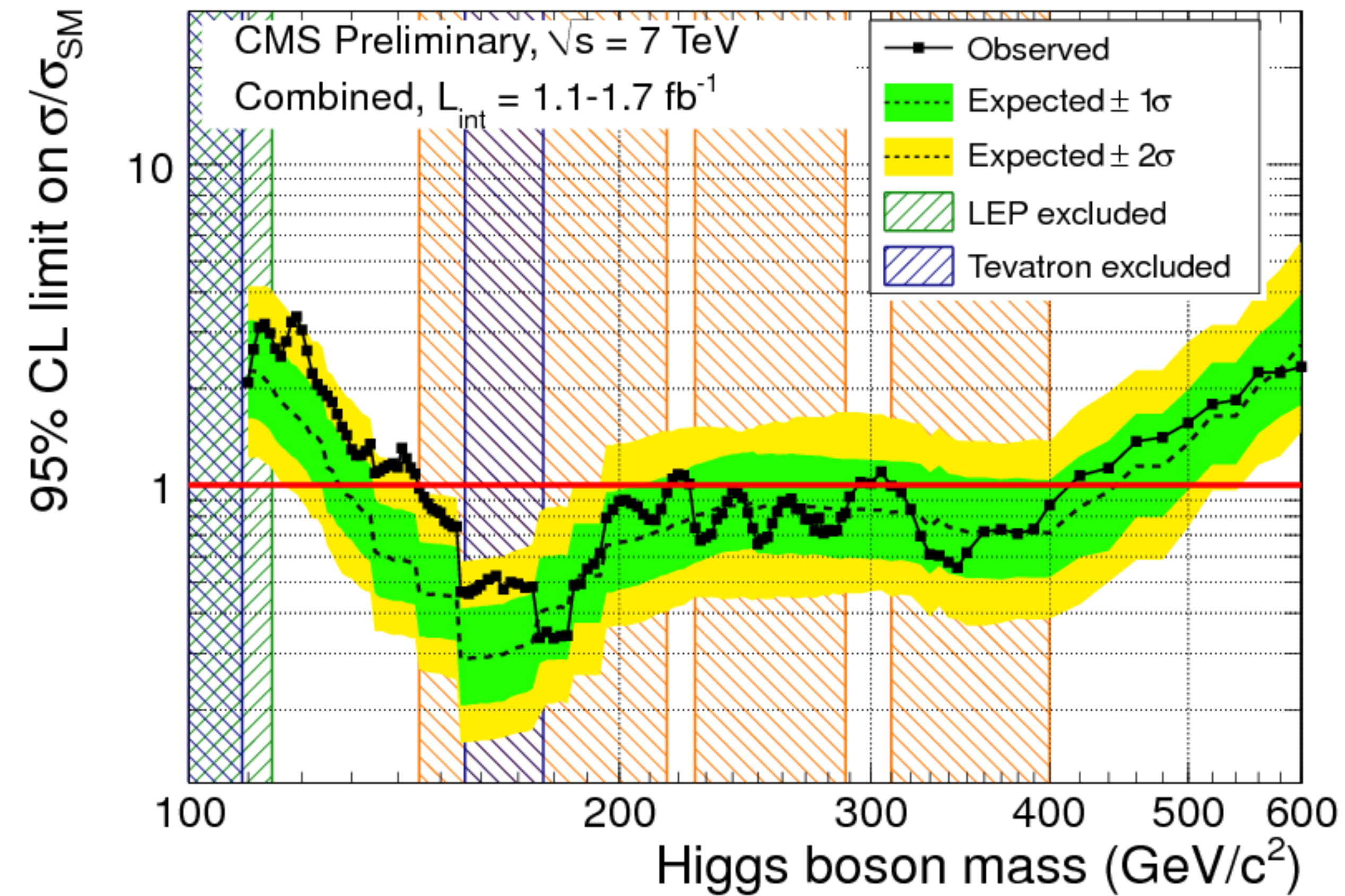


- Total of 27 events are selected in data: 6 (4e), 9 (2e2μ), 12 (4μ)
- Expected: 28±4

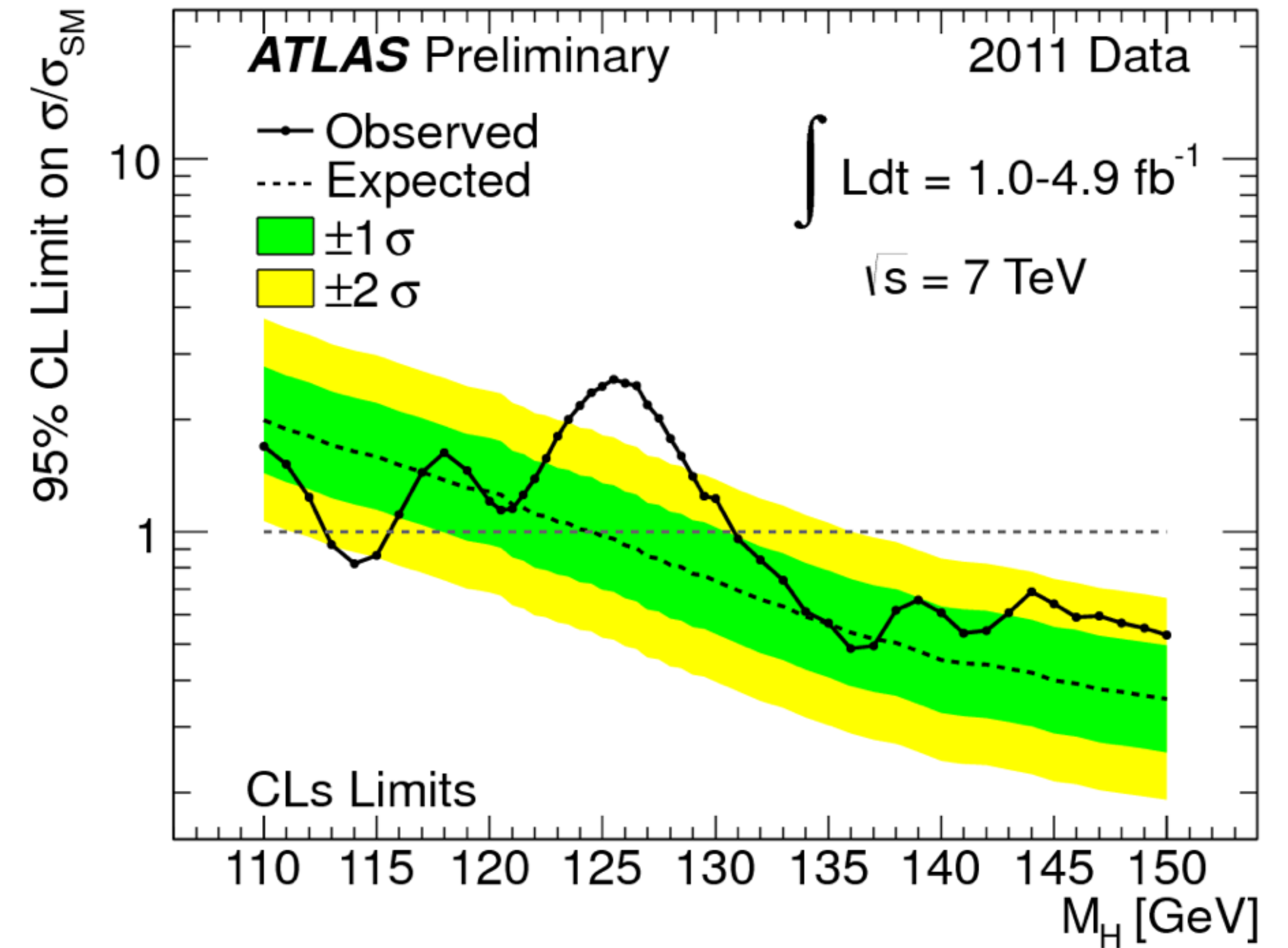
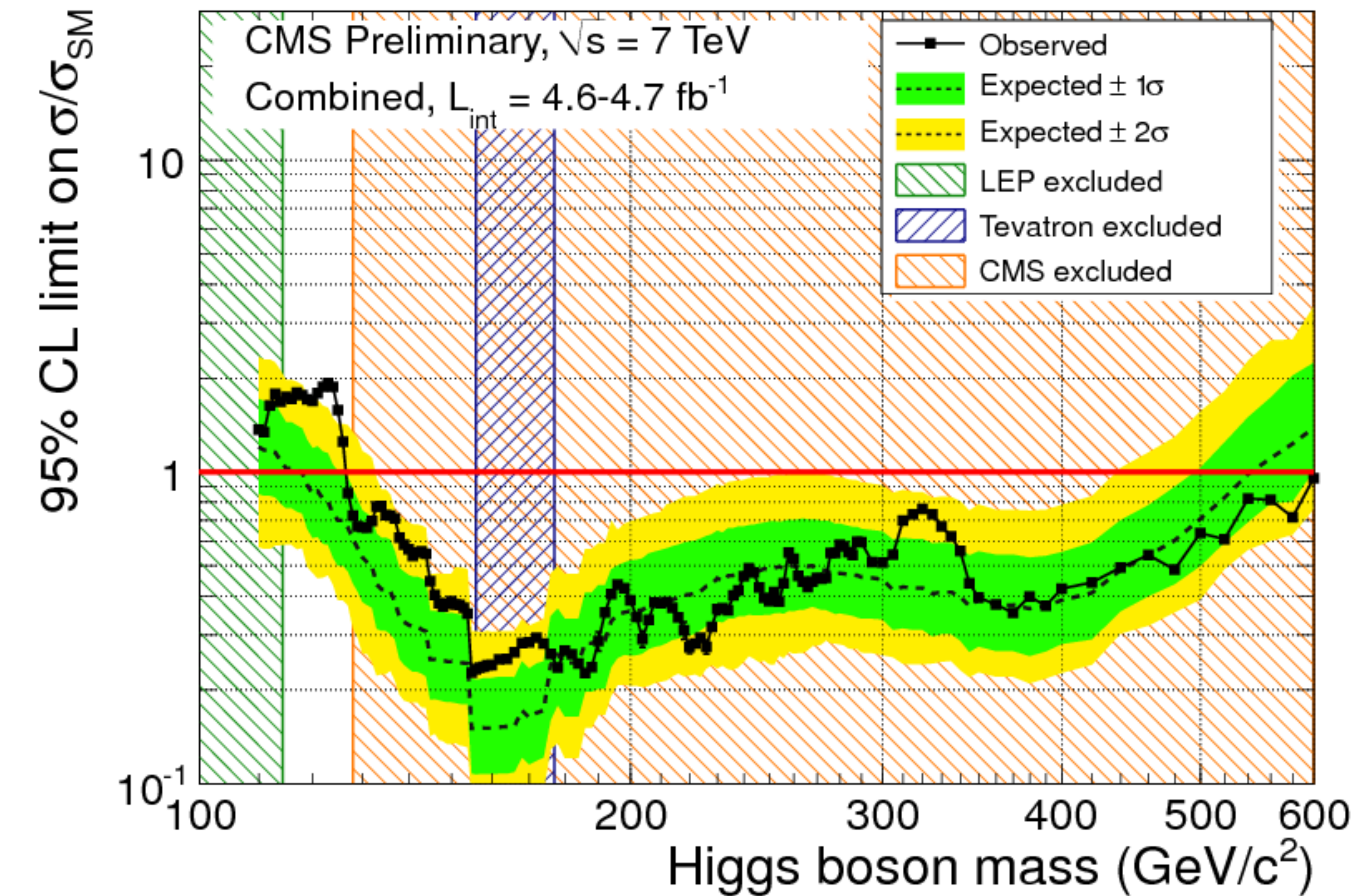


	baseline		
	4e	4μ	2e2μ
ZZ	4.05 ± 0.26	6.02 ± 0.40	9.87 ± 0.66
Z+jet	0.48 ± 0.08	0.09 ± 0.02	0.61 ± 0.11
Zb $\bar{b}$ /c $\bar{c}$, t $\bar{t}$	0.01 ± 0.01	0.05 ± 0.01	0.06 ± 0.01
WZ	0.009 ± 0.009	0.009 ± 0.009	0.04 ± 0.02
All background	4.54 ± 0.27	6.12 ± 0.40	10.52 ± 0.67
$m_H = 140 \text{ GeV}/c^2$	0.45	0.82	1.19
$m_H = 200 \text{ GeV}/c^2$	1.20	1.71	2.80
$m_H = 350 \text{ GeV}/c^2$	0.70	0.93	1.63
Observed	5	10	6

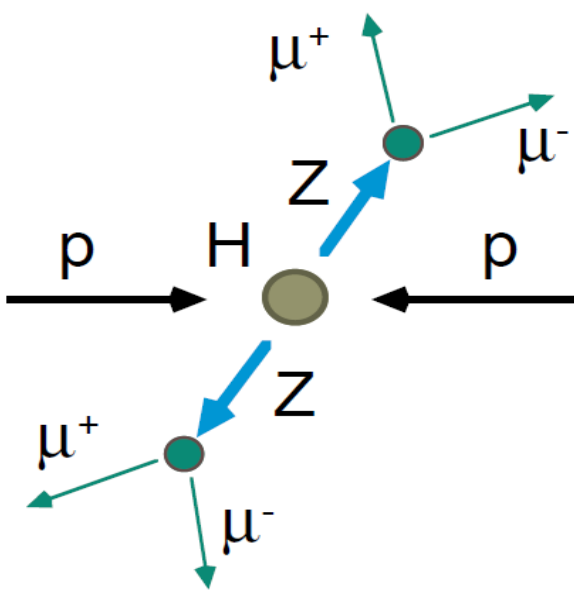
Higgs, Summer 2011



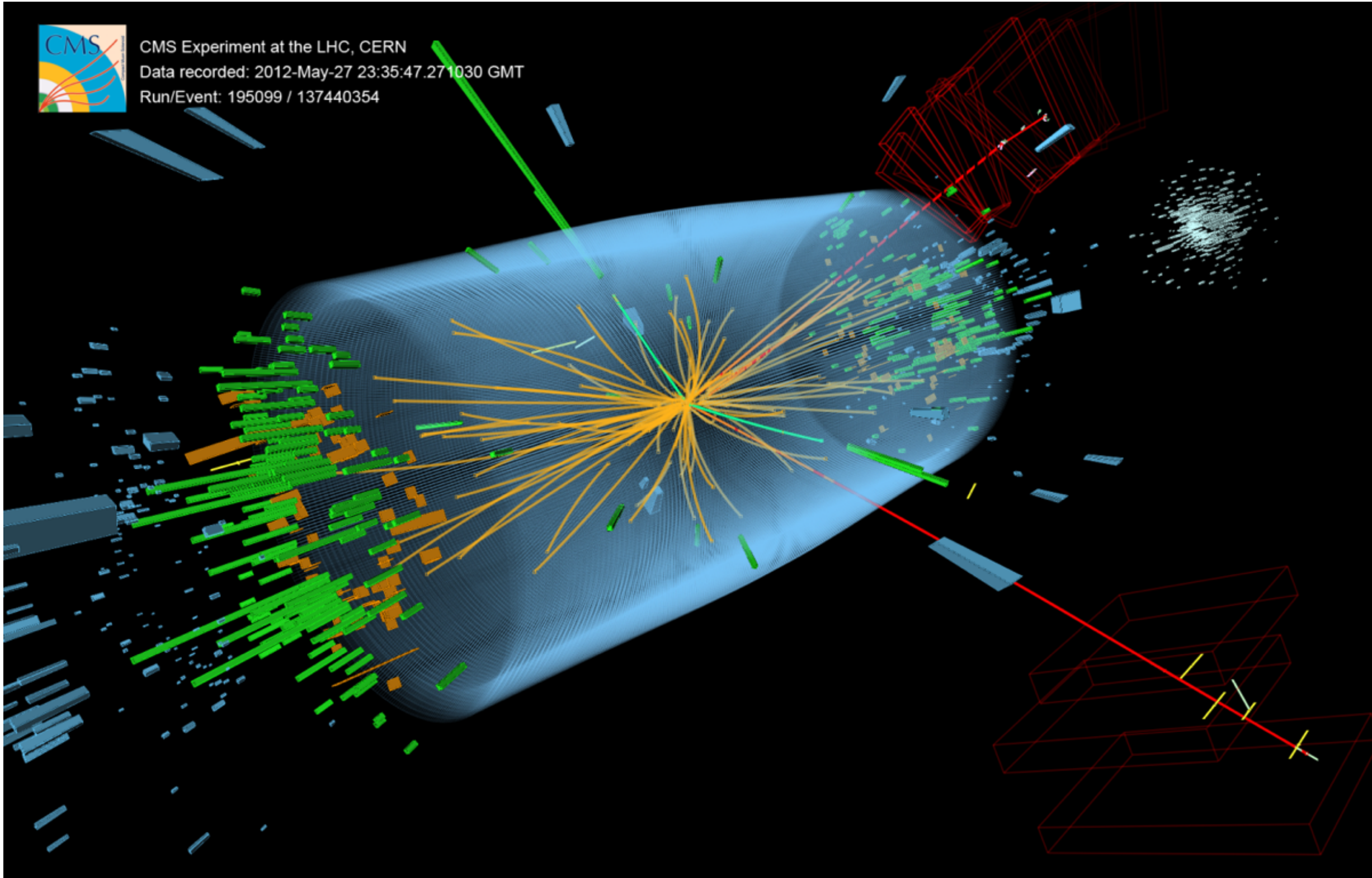
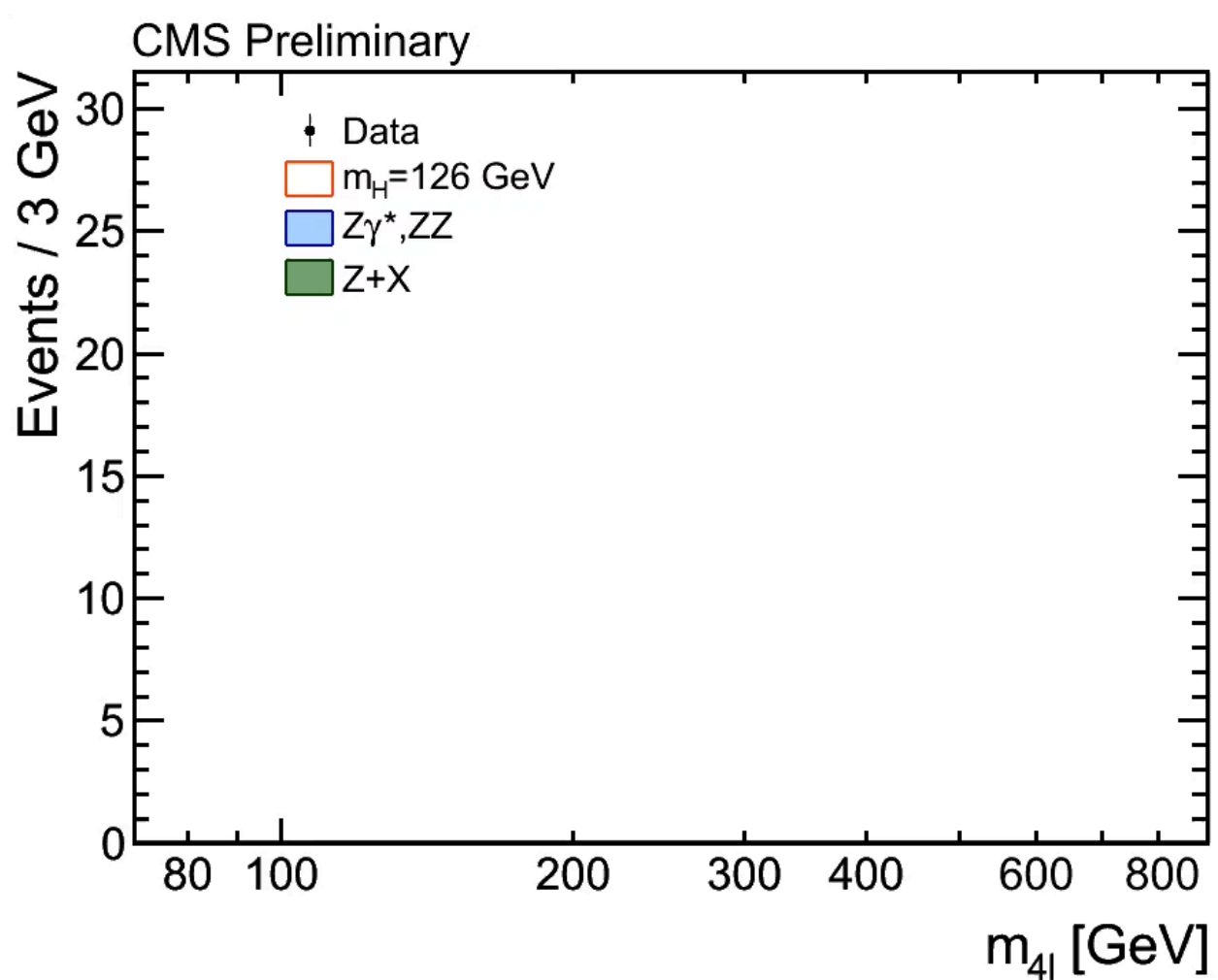
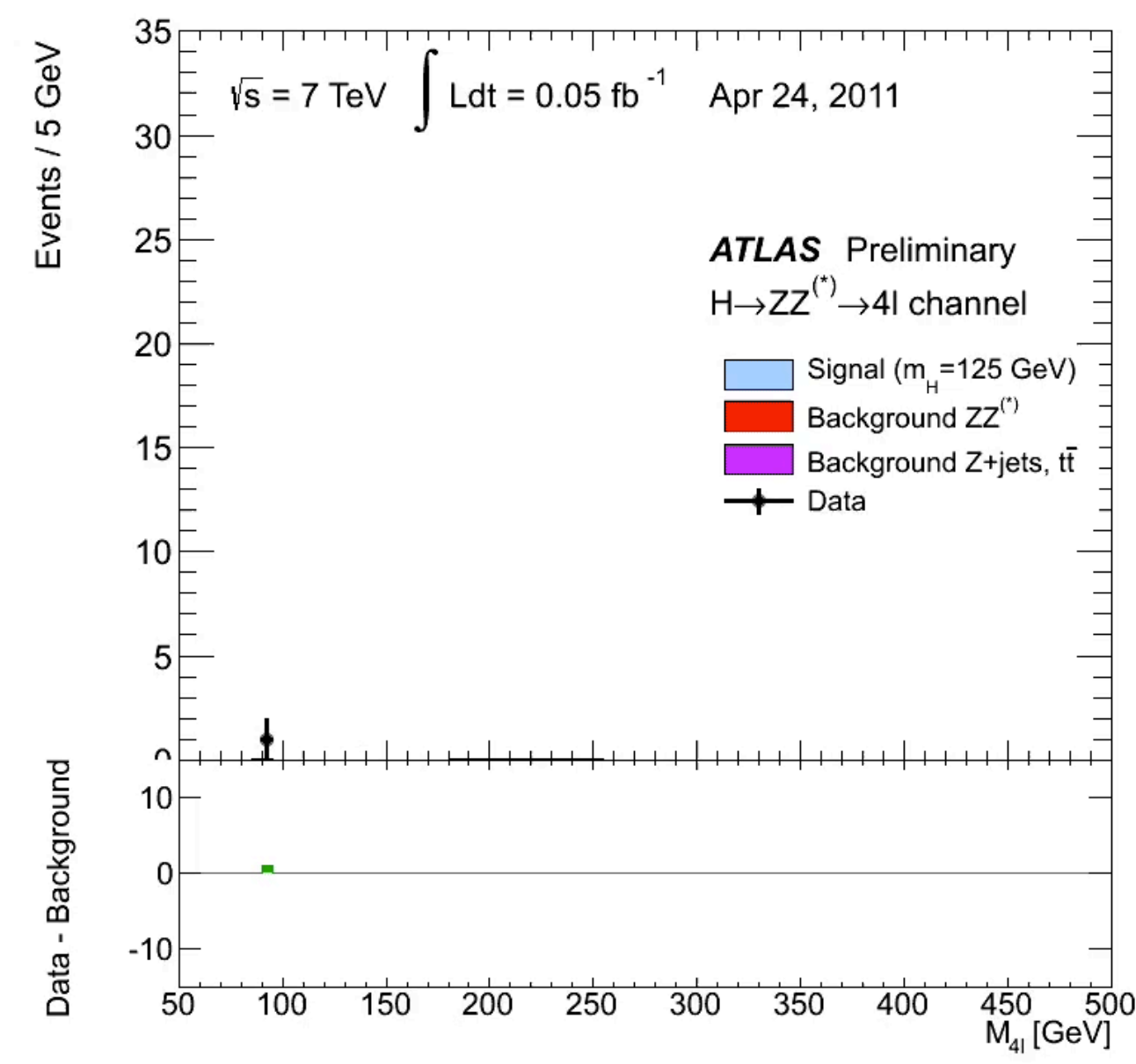
Higgs, December 2011



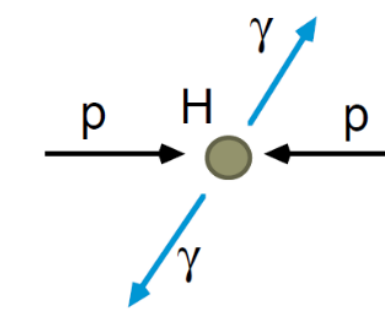
Higgs decay to ZZ (leptons)



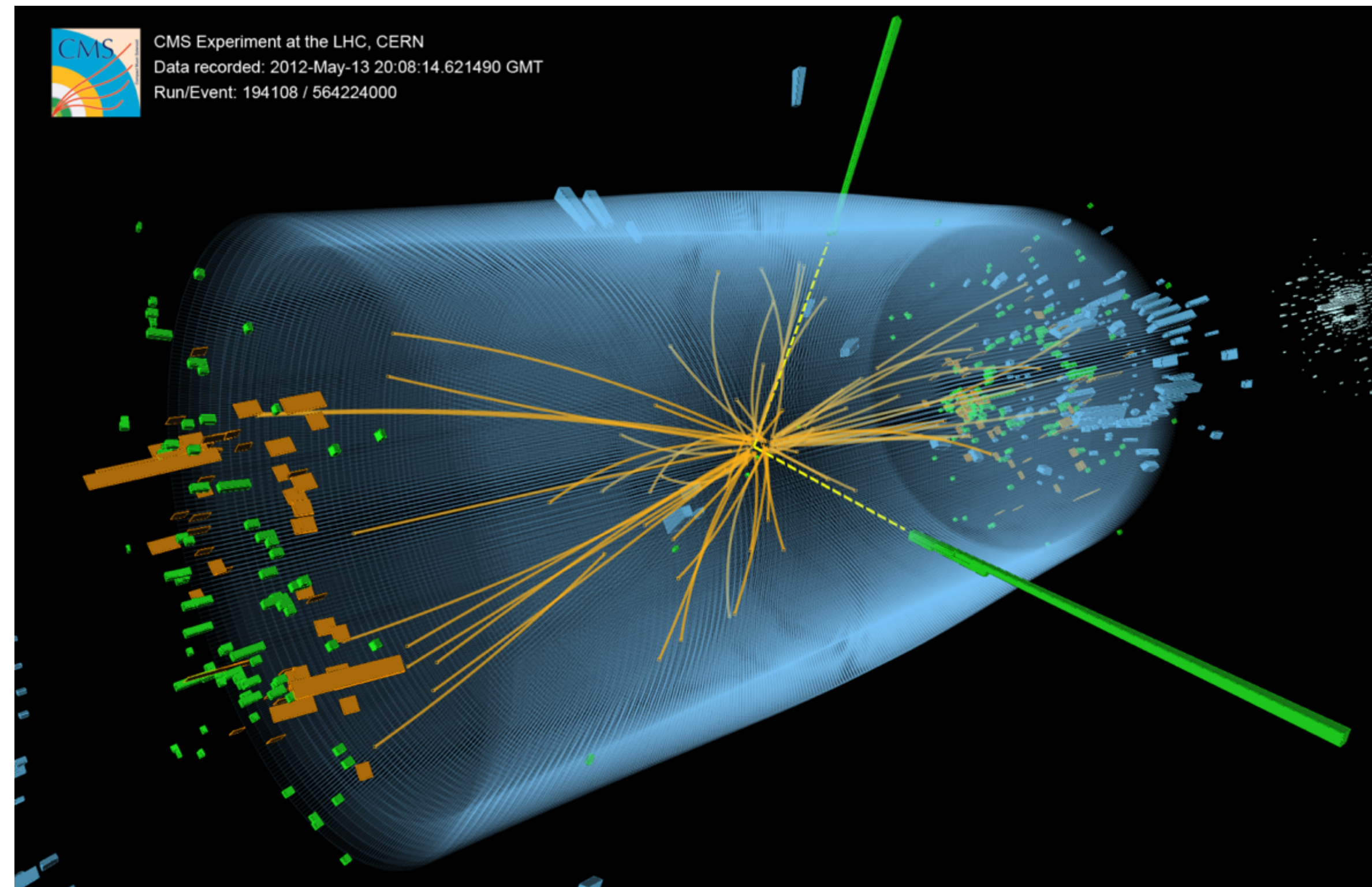
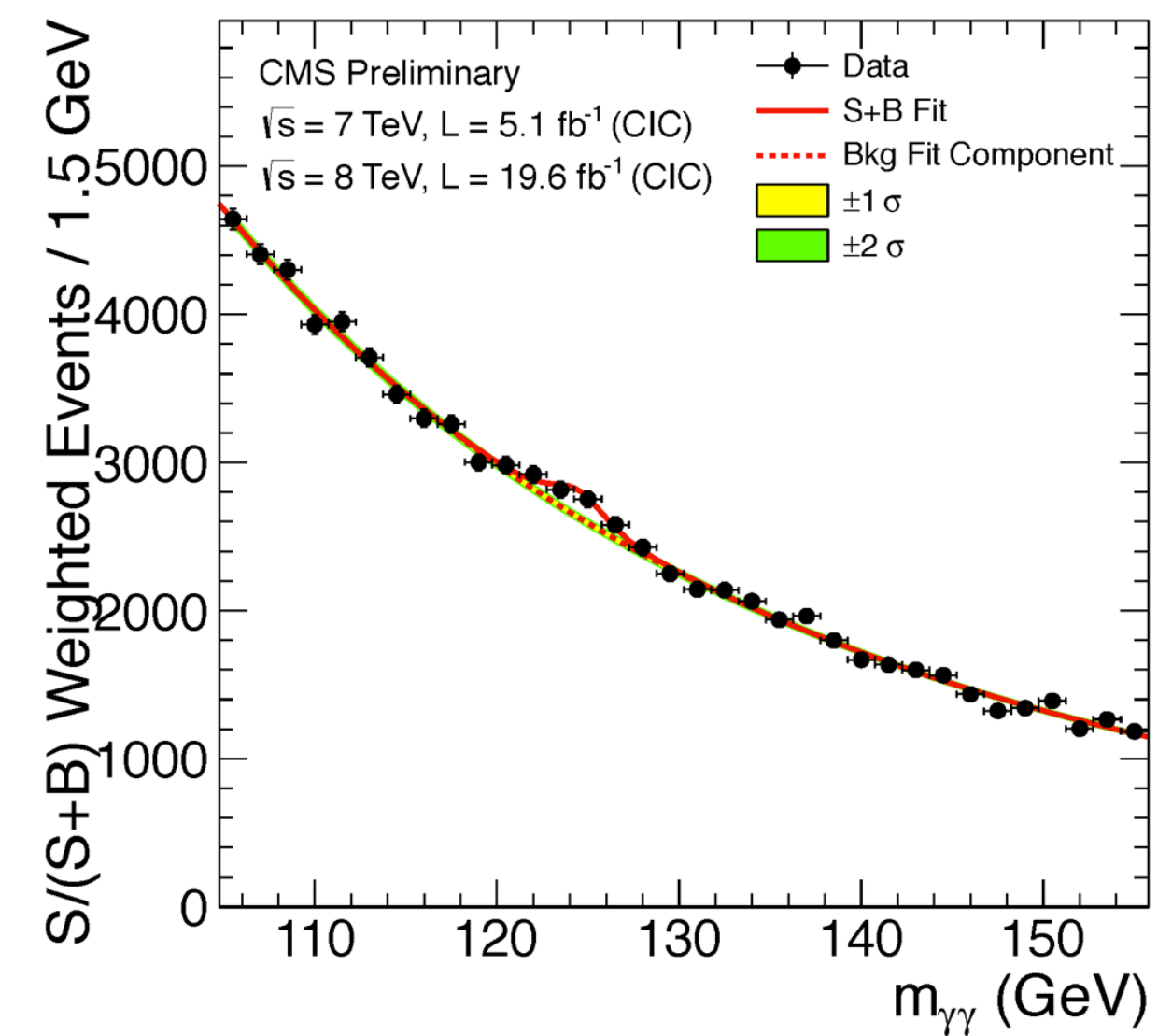
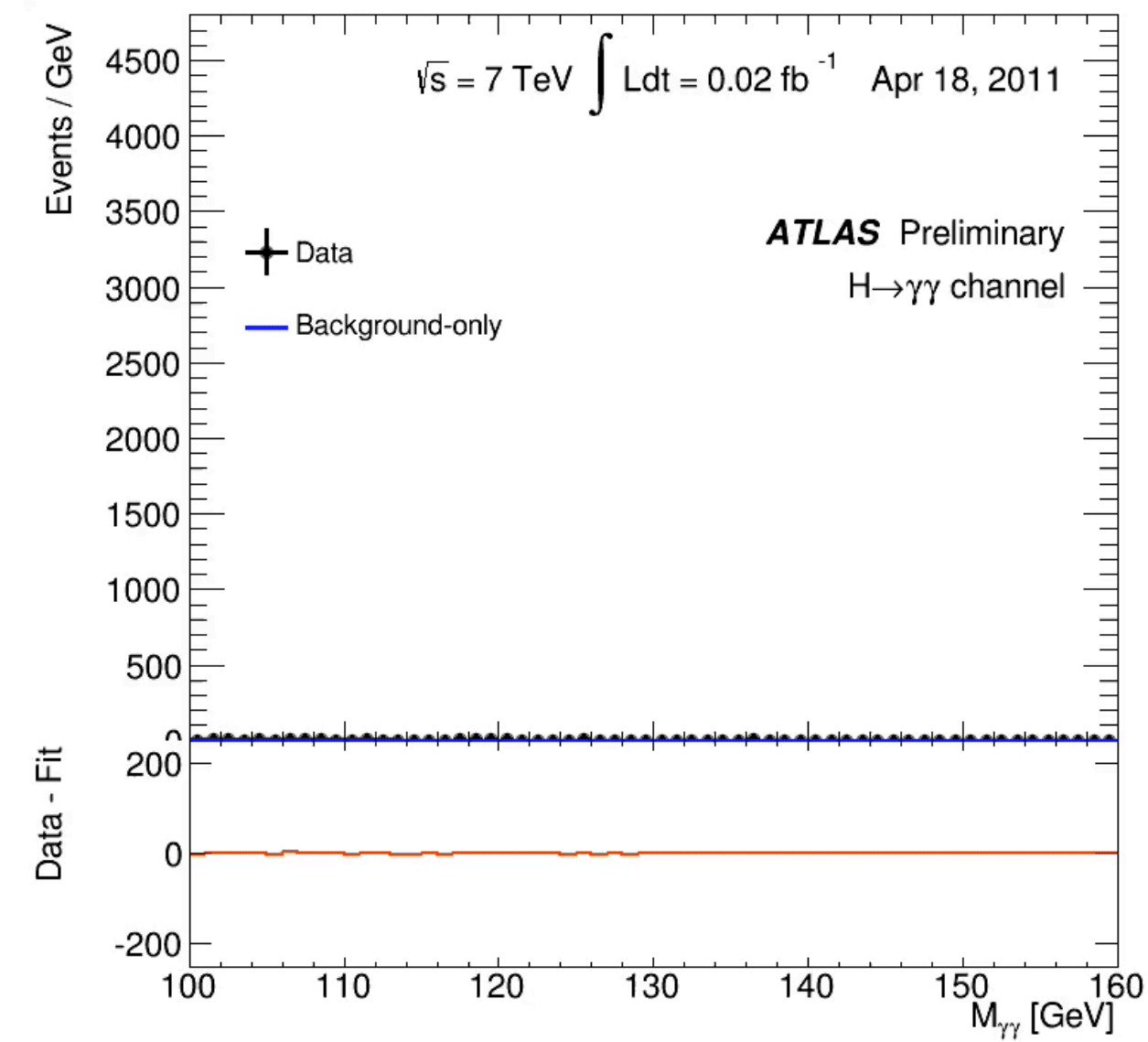
$$mass = \sqrt{\vec{p}_1^2 + \vec{p}_2^2 + \vec{p}_3^2 + \vec{p}_4^2}$$



Higgs decay to Photons

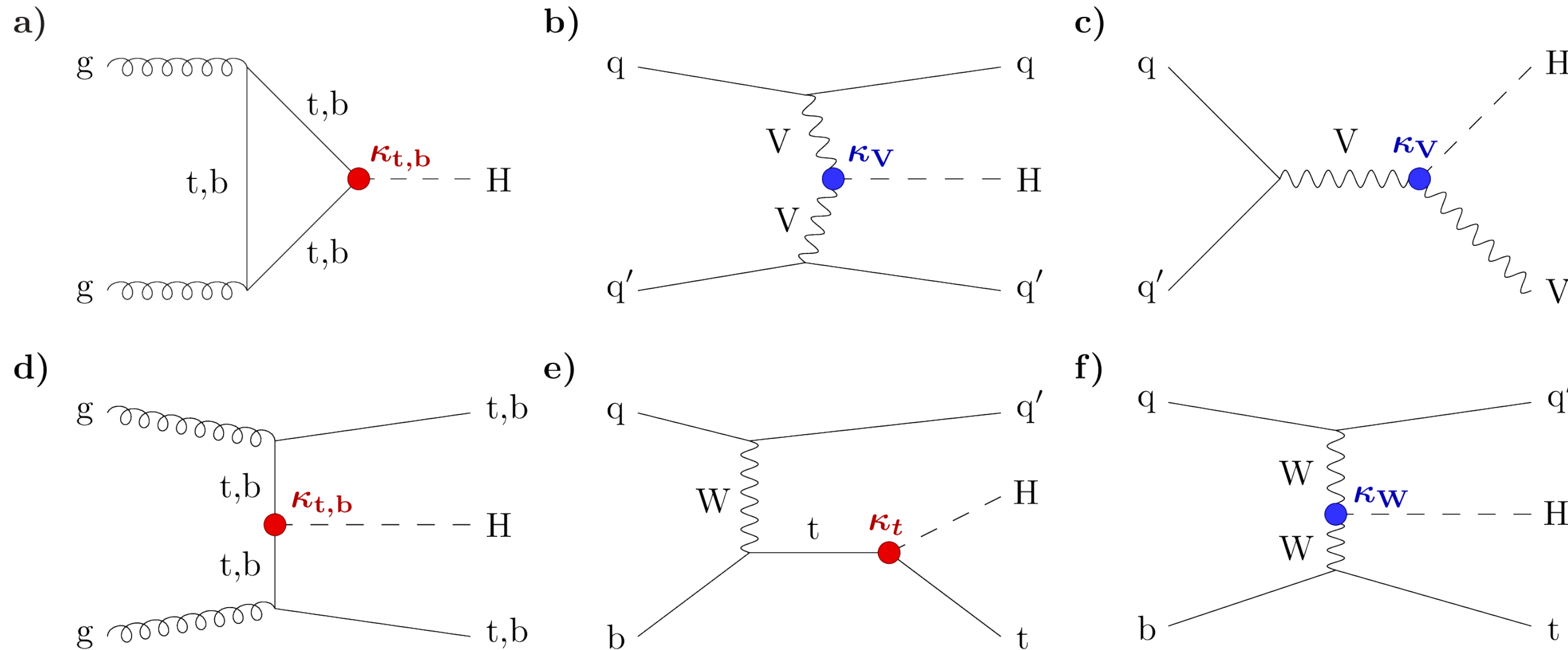


$$mass = \sqrt{2E_{\gamma_1}E_{\gamma_2}(1 - \cos \theta_{12})}$$

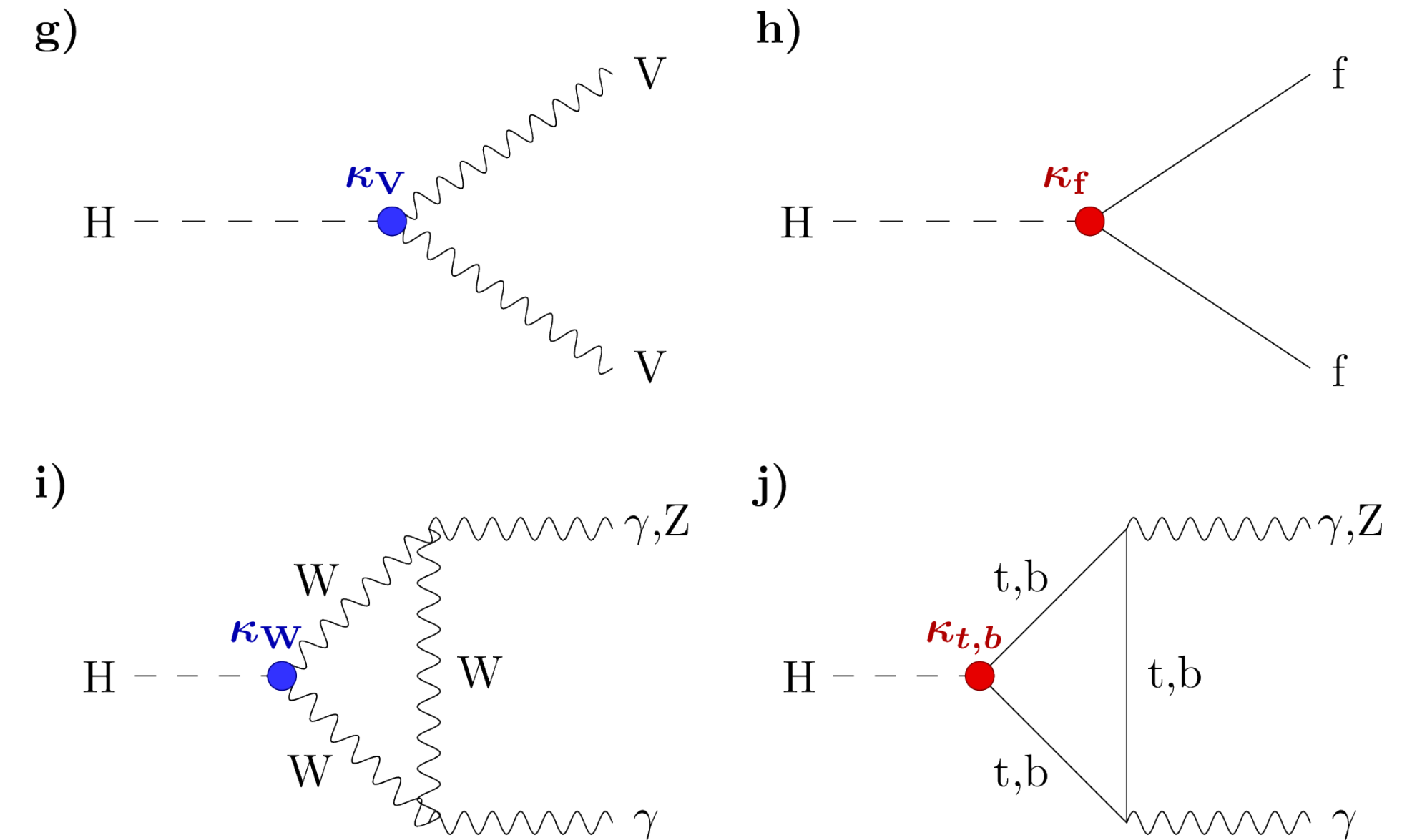


Higgs today

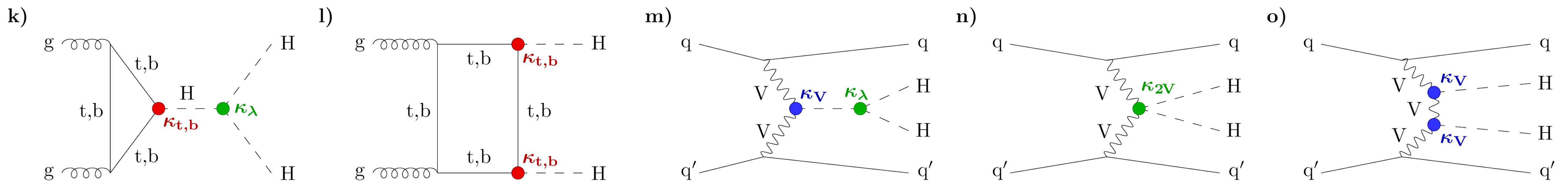
Higgs boson production modes



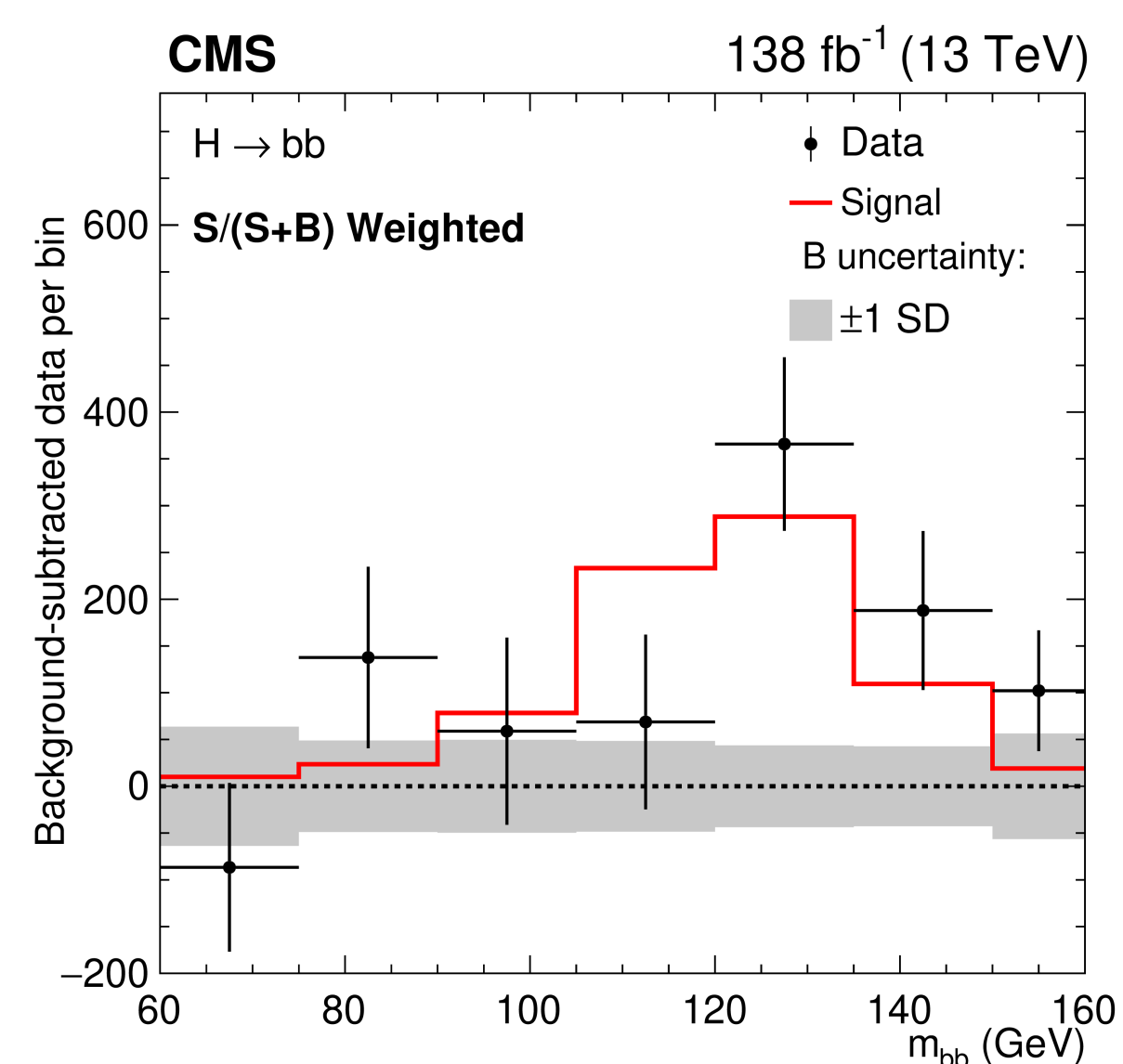
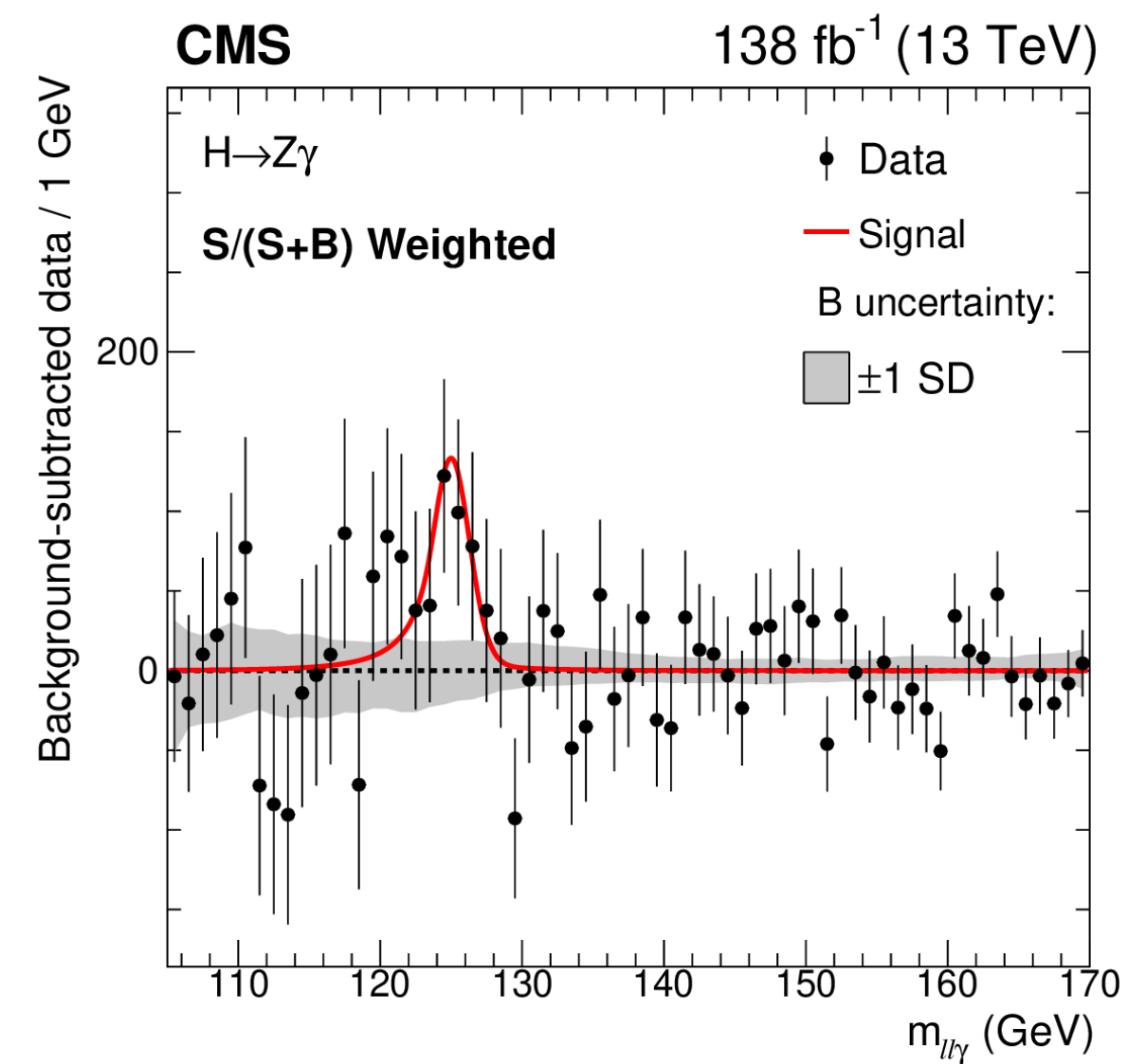
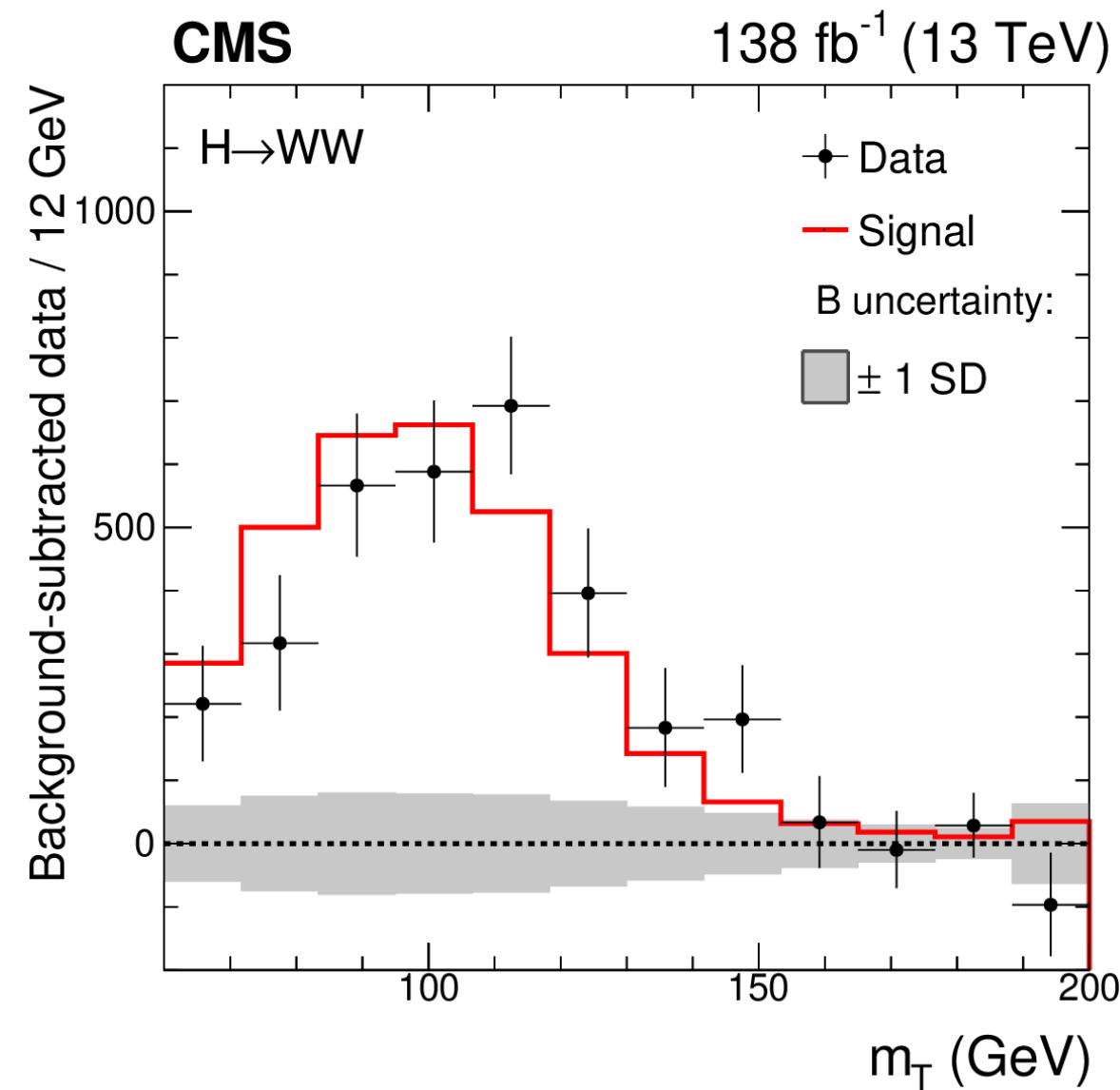
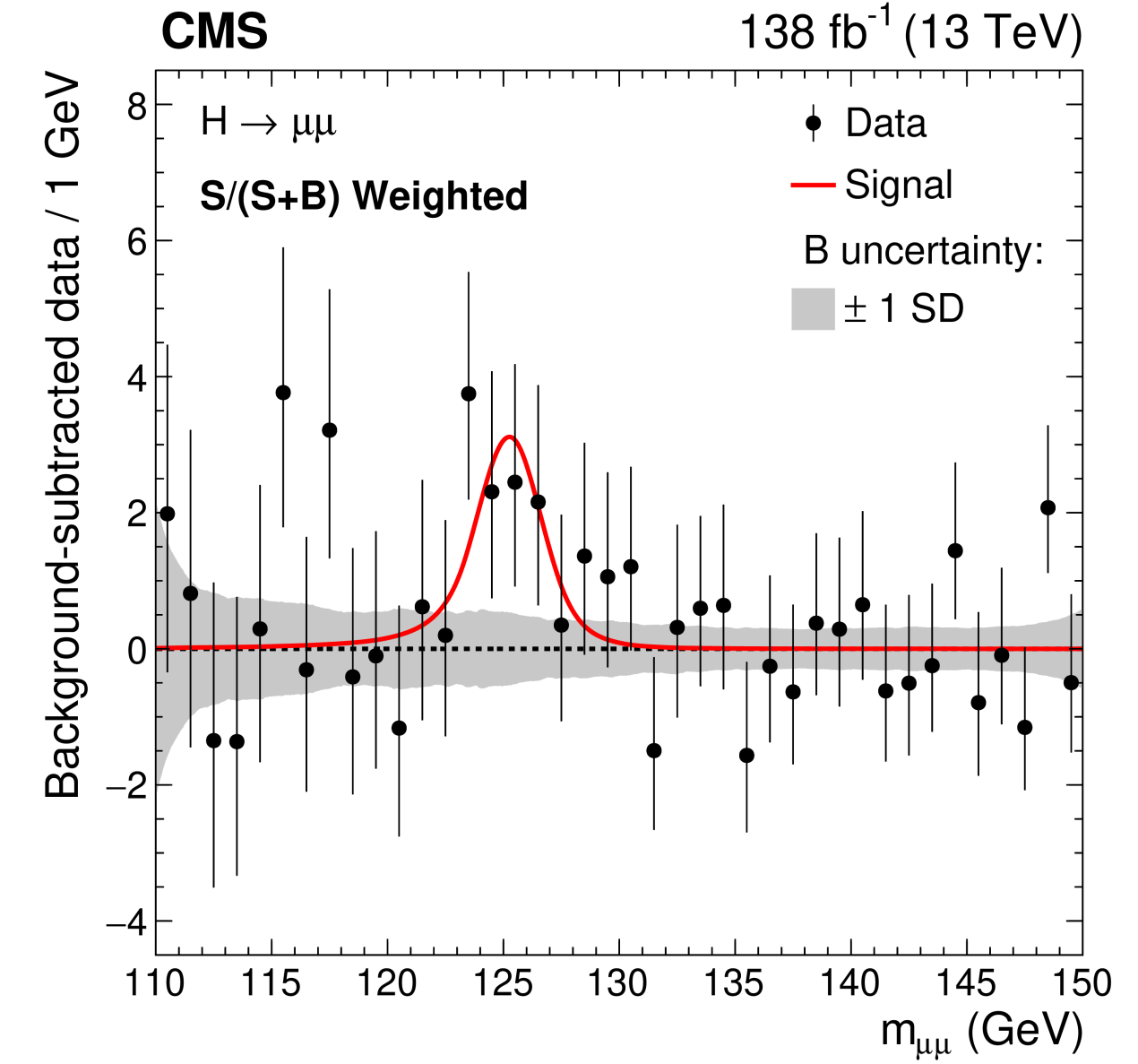
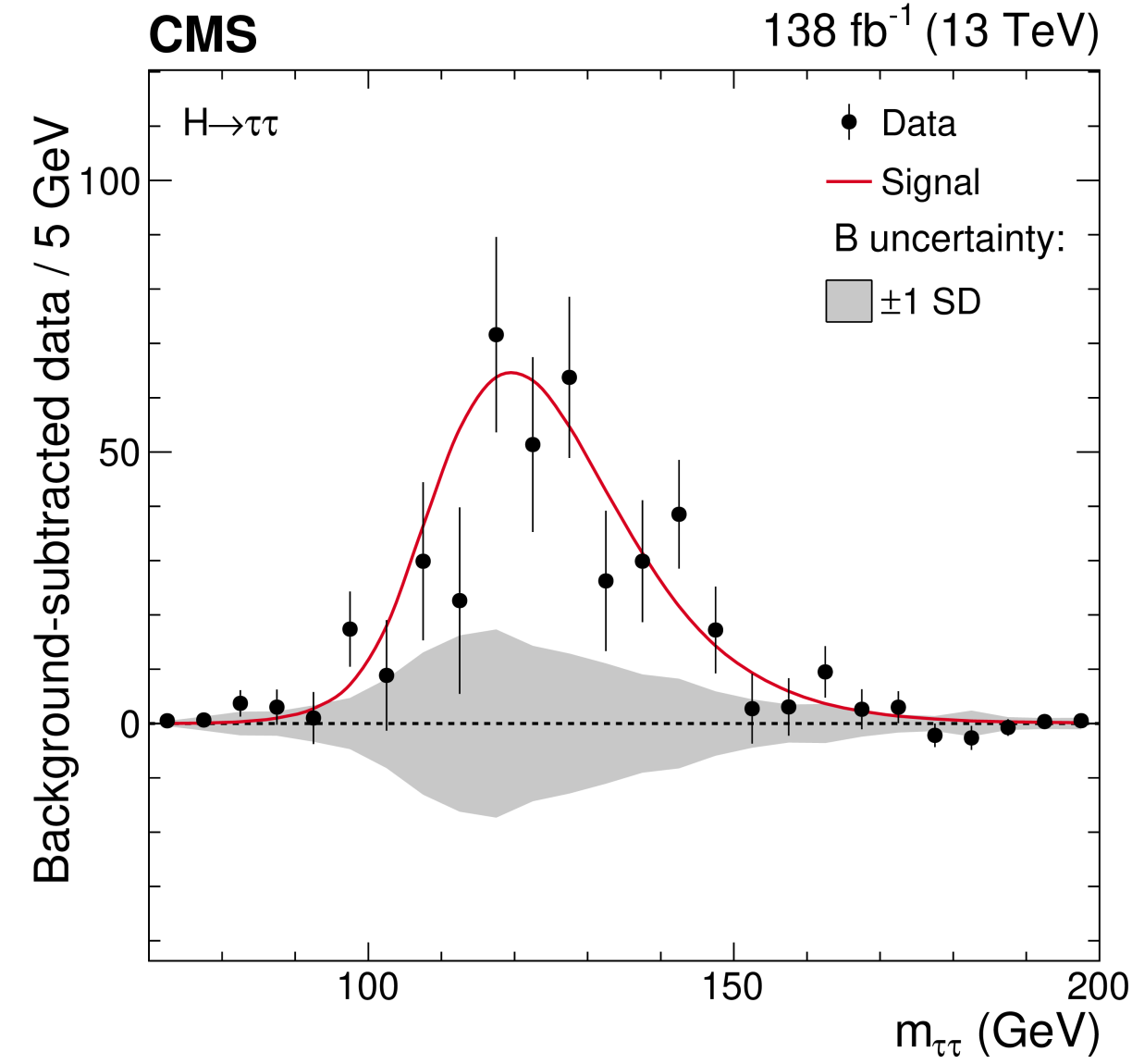
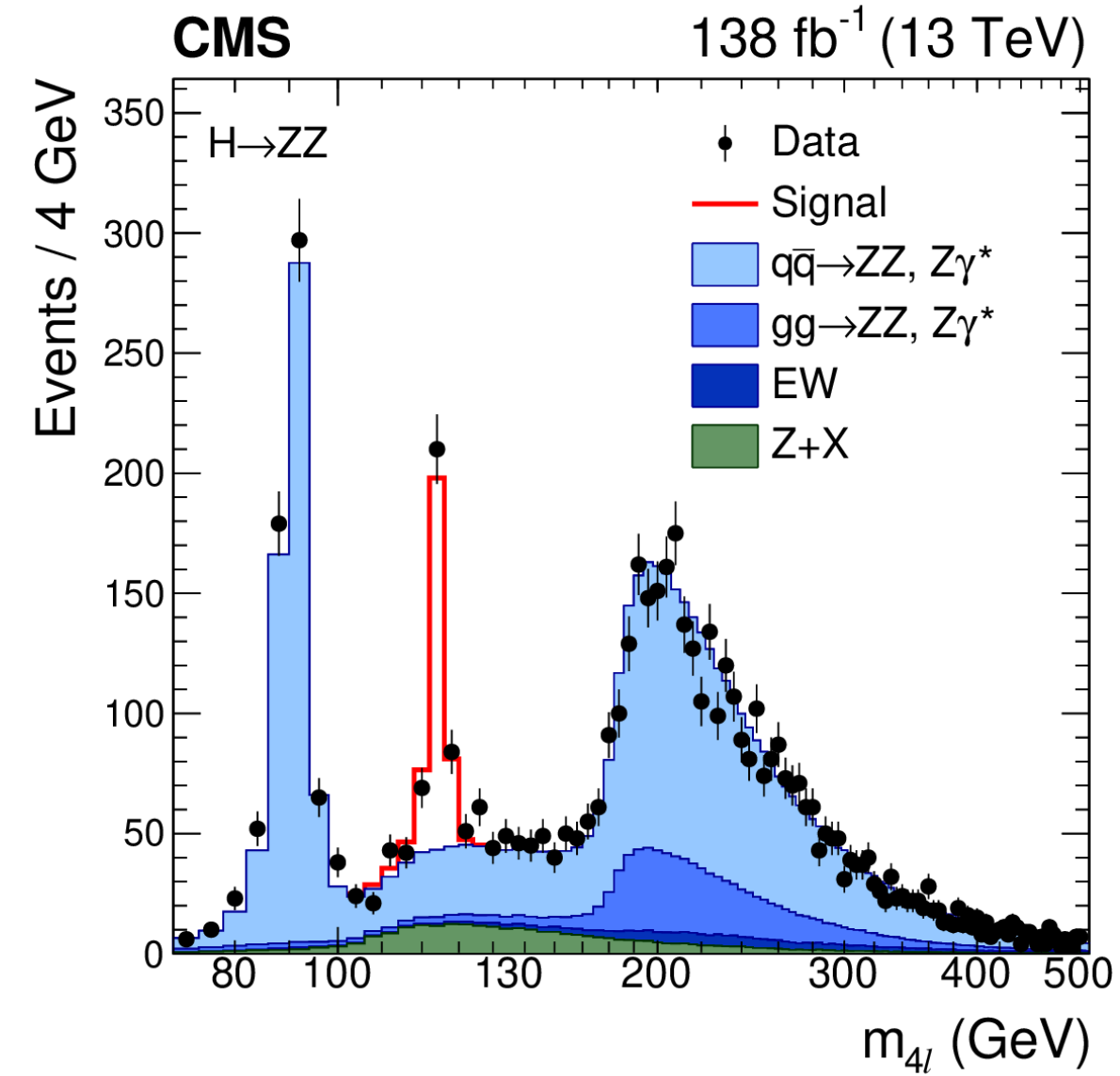
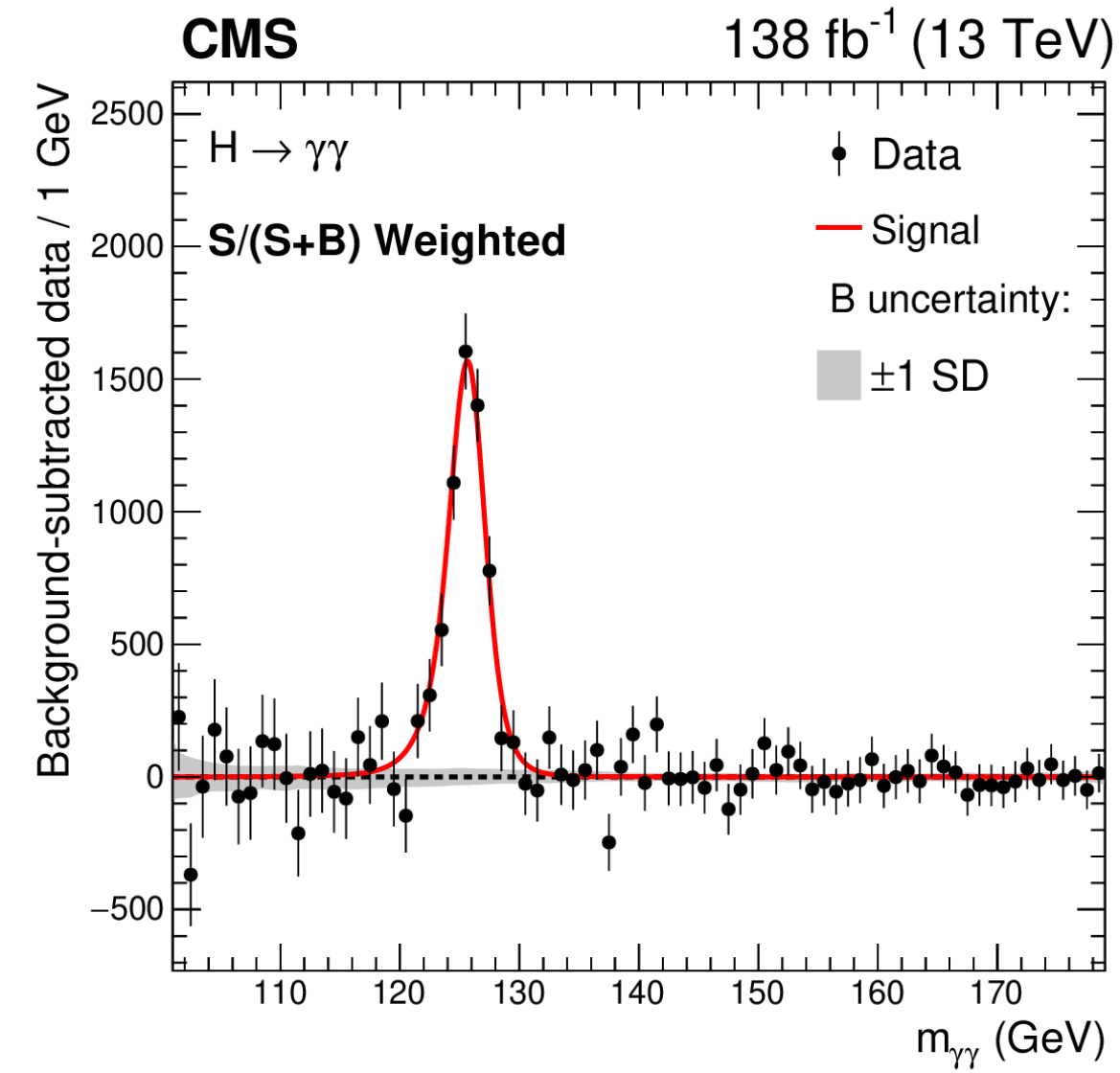
Higgs boson decay channels



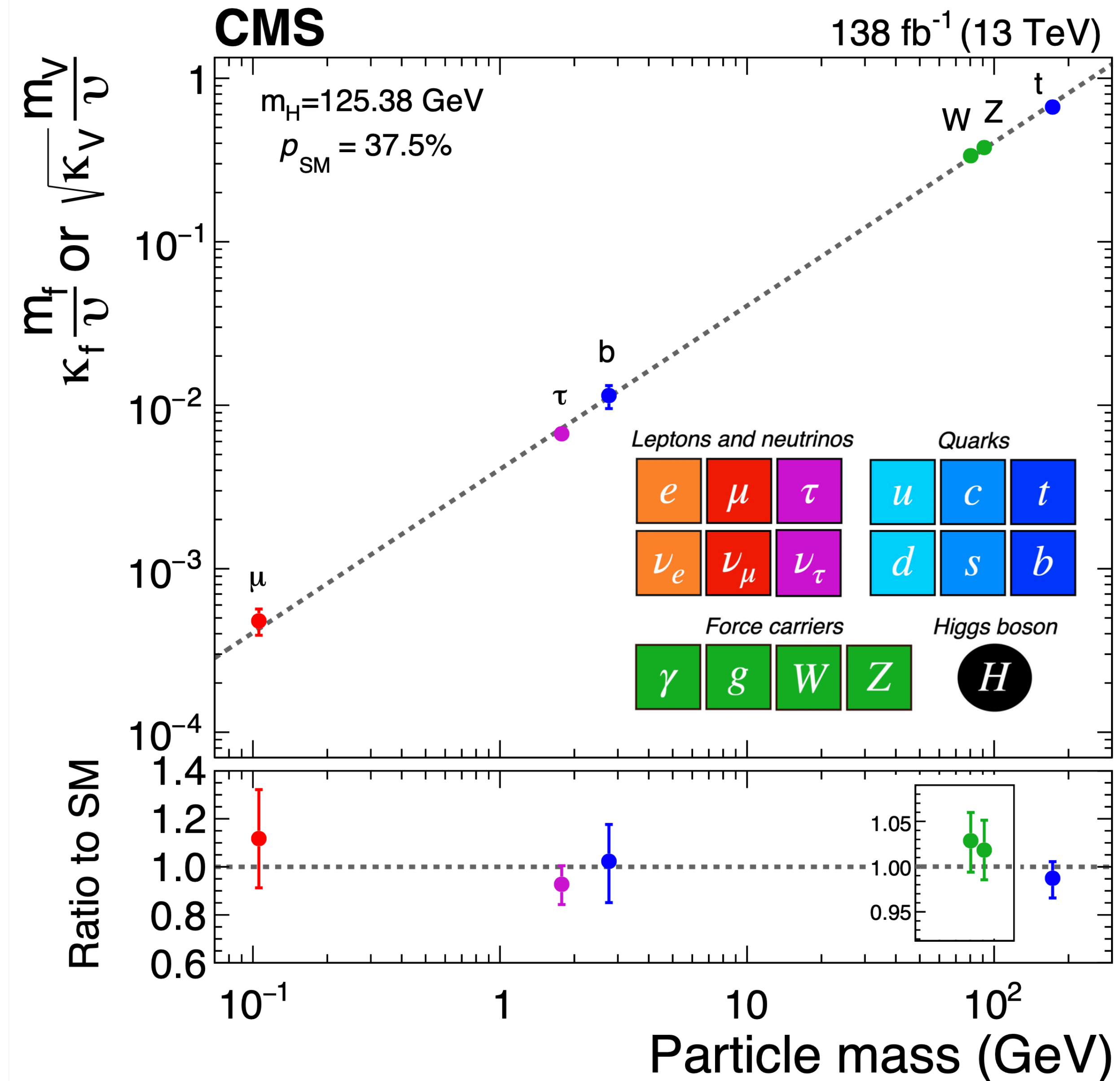
Higgs boson pair production



Higgs after Run2



Higgs couplings today



Energy Frontier after Higgs Discovery

- ▷ Intense scrutiny of Higgs and Yukawa sector

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}D\psi$$

Precision Electroweak and QCD

$$+|D_{\mu}\phi|^2 - V(H)$$

Higgs properties
Higgs self interaction

$$+Y_{ij}\psi_i\psi_j\phi + \text{h.c.}$$

Higgs coupling to bosons and fermions
CKM matrix and CP Violation

- ▷ While keeping a wide open eye on new phenomena

$$+\mathcal{L}_{\text{New}}$$

New light and heavy particles
Lepton flavour universality violation
Leptoquarks
SUSY
Long-lived particles
Dark matter